

Physics-informed networks for track extrapolation

From van der Pol to the LHCb magnet

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A label-free surrogate for the field-propagation part of the LHCb track extrapolator,
built from the discrete-time construction and validated first on an oscillator.

Where this talk goes

1. **The problem** (3 slides). What the extrapolator does, why a fixed arithmetic cost is the point, and the two questions I set in advance with their pre-stated meanings.
2. **Theory** (10 slides). The equation of motion in z ; Gauss–Legendre collocation; the reconstruction residuals and the loss; the scales; the straight-line-residual parameterisation; the continuous-time objective; the splice construction and its $1/T$ price; pseudo-time stepping; the comparators and metrics.
3. **The fix literature** (4 slides), one paper per slide: what it proves, its protocol, my implementation check, my result.
4. **System and data** (5 slides). Why van der Pol first, the shared machinery, the dictionary between the two problems, and both data sets.
5. **Results on van der Pol** (10 slides). The cliff, causal weighting, the supervised control, eight objectives to convergence, the failure classes, the two resource axes, the measured $1/T$ price, and the whole discrete-time column.
6. **Results on LHCb** (13 slides). Six experiments with their full tables, then the assembled verdict and the cost statement.
7. **What is not working** (2 slides), and **what would falsify the edge** (1 slide) with pass and fail criteria stated in advance.
8. **Direction** (2 slides) and **what I would like feedback on**.

Provenance, the harness and the links to the full record are at the end.

The extrapolator transports one track state between two planes, millions of times a second

A **track state** is five numbers at one value of z :

$$S = (x, y, t_x, t_y, qop), \quad qop = 0.299792458 \frac{q}{p[\text{GeV}]},$$

positions in millimetres, slopes $t_x = dx/dz$, $t_y = dy/dz$ dimensionless, and the Allen convention for charge over momentum.

The **extrapolator** maps $S(z_0) \mapsto S(z_1)$. It is called by the Kalman fit at every measurement, by the matching of tracks found in front of and behind the magnet, and by every vertex fit that transports tracks back towards the collision point.

The dipole peaks near 1.05 T around $z = 4.7$ m. On the crossing studied here the true path departs from a straight line by a median of **520 mm**, so the quantity to get right is a half-metre correction, and getting it right to $100 \mu\text{m}$ is one part in five thousand.

What the standard implementation does

- Steps along z in small increments, evaluating the field map at each step. Accurate. Its cost is neither fixed nor predictable, because the trip count depends on the track.

Scope, fixed at the start

- The surrogate replaces **field propagation only**. Multiple scattering off detector material and energy loss stay with the existing extrapolator.
- The size of what I am excluding is measured, not assumed: the label disagrees with the particle's own next simulated hit by a median of $22 \mu\text{m}$, $6.3 \mu\text{m}$ above 5 GeV and $86 \mu\text{m}$ below 2 GeV. That $1/p$ behaviour is the material effect.
- So a surrogate that reached the $23 \mu\text{m}$ scheme ceiling would be at the level where material starts to matter more than propagation.

A cost that is fixed in advance is worth more than a cost that is merely small

Solving the eight-stage scheme directly

- A root-finder iterates on a coupled nonlinear system of $q + 1$ vector equations.
- Median **44 residual evaluations** per leg on the magnet crossing, each of them eight field lookups plus the root-finder's internal Jacobian work.
- The trip count depends on the track and is not known before the loop starts.

Asking a trained network instead

- **One forward pass** emitting all eight stage states and the endpoint at once: $9 \times 4 = 36$ numbers, eight field lookups, two small matrix products.
- No iteration and no branch, so the arithmetic is identical for every track in the batch.
- A trigger running on graphics processing units can schedule a fixed cost. It cannot easily schedule a loop whose length depends on the input.

Stated plainly, because it is the weakest link in the argument: 44 against 1 is an *operation count*, not a measurement. Nothing in this talk is a throughput benchmark against the production extrapolator. Everything here runs in double precision on one processor core, because that is what makes the physics comparisons clean, and that is not where the surrogate would live.

The goal in one sentence, and the two questions I wrote down first

A network that turns a starting state into a trajectory,
trained from the equations alone, with no correct answer anywhere in the loss.

Question 1: does it work?

Pre-stated meaning. A yes requires all three:

- the label-free training converges at all;
- where the numerical scheme is the limiting factor, the network's error sits on the scheme's own error;
- where the network is the limiting factor, the shortfall is attributed to something identified rather than left as a mystery.

A question about the *method*.

Question 2: can it be used?

Pre-stated meaning. A yes requires all three:

- one network serves steps of arbitrary length starting anywhere;
- it beats the trivial baseline of ignoring the magnet on every kind of step;
- its error does not compound uncontrollably when the network is applied to its own output.

A question about the *product*. They are separable, and they came out differently.

The source paper contains two constructions, and they differ in kind

M. Raissi, P. Perdikaris and G. E. Karniadakis, J. Comput. Phys. 378 (2019) 686

Continuous time, their section 3.1

The network *is* the solution: t in, state out. The residual $r = \dot{y}_\theta - f(y_\theta)$ is formed by automatic differentiation and driven to zero at sampled collocation points, with a single anchor at $t = 0$.

Demonstrated on Burgers on $t \in [0, 1]$ and Schrödinger on $t \in [0, \pi/2]$, at 6.7×10^{-4} and 1.97×10^{-3} . Sub-period horizons in every case.

No fixed arithmetic cost per step. A weak constraint per training point. **And, as the next slides show, an objective that accepts an infinite family of wrong answers.**

Discrete time, their section 3.2

The network represents *one step* of an implicit Runge–Kutta scheme: given a state, it emits all q Gauss–Legendre stage states and the endpoint at once, and the algebra of the step supplies the supervision.

Demonstrated on Burgers and Allen–Cahn, in both cases as a single very large time step with a high-stage scheme, and reported to tolerate step sizes far beyond conventional stability limits.

Its stated limitation is the one I attack: the paper trains a network for one specific initial state and one specific Δt , and says nothing about reuse. I make the starting state, the start plane and the step length inputs.

The structural difference that decided the order of work: in the discrete-time construction the **known starting state enters the loss as data** through every one of the $q + 1$ equations. There is no interval over which a hand-over can be hidden, and no free phase. A constant output is not a minimiser.

The equation of motion in z , and its three awkward properties

Newton's second law with the Lorentz force, rewritten with z as the independent variable and in the state variables above:

$$\begin{aligned}\frac{dx}{dz} &= t_x, & \frac{dy}{dz} &= t_y, \\ \frac{dt_x}{dz} &= \kappa \text{qop } N [t_x t_y B_x - (1 + t_x^2) B_y + t_y B_z], \\ \frac{dt_y}{dz} &= \kappa \text{qop } N [(1 + t_y^2) B_x - t_x t_y B_y - t_x B_z], \\ \frac{d\text{qop}}{dz} &= 0,\end{aligned}$$

with $\kappa = 10^{-3}$ the unit constant matching the qop convention,
 $N = \sqrt{1 + t_x^2 + t_y^2}$ the path-length to z -length ratio, and B in tesla at the point the particle currently occupies.

- **Non-autonomous.** The right-hand side depends on z explicitly through the field. Unlike every textbook problem the technique was demonstrated on.
- **qop is conserved.** A magnetic field does no work, so momentum is an input carried through unchanged and only four components are ever predicted.
- **The field is not an analytic function.** It is a measured table on an $81 \times 81 \times 146$ grid at 100 mm spacing, read by trilinear interpolation. The interpolant is C^0 and not C^1 : the gradient jumps at every cell boundary.

Sanity check. $p = 10$ GeV, $q = +1$, so $\text{qop} = 0.0300$; $t_x = t_y = 0$, $B_y = -1.0$ T. Then $dt_x/dz = 3.0 \times 10^{-5}$ per mm, and over 5178 mm with a constant field the sideways displacement is about 400 mm. Measured median: 520 mm.

Implicit Runge–Kutta is collocation: a degree- q polynomial that solves the equation at q nodes

$$S_j = S_0 + \Delta z \sum_{k=1}^q A_{jk} f(S_k, z_0 + c_k \Delta z), \quad j = 1, \dots, q,$$
$$S_{\text{end}} = S_0 + \Delta z \sum_{k=1}^q b_k f(S_k, z_0 + c_k \Delta z).$$

(c, A, b) is the tableau: nodes, stage matrix, weights. A strictly lower triangular gives an explicit method; A full gives an **implicit** one, and the stage equations become a coupled nonlinear system.

Gauss–Legendre. Nodes are the roots of the degree- q Legendre polynomial mapped onto $[0, 1]$; A and b integrate polynomials exactly to the highest possible degree. Formal order $2q$, the maximum for q stages, and equivalent to polynomial collocation: the stage states are the values at the nodes of the degree- q polynomial that satisfies the equation exactly at those nodes.

The $q = 2$ tableau, worked

$$c = \left(\frac{1}{2} - \frac{\sqrt{3}}{6}, \frac{1}{2} + \frac{\sqrt{3}}{6} \right) \approx (0.2113, 0.7887), \quad b = \left(\frac{1}{2}, \frac{1}{2} \right),$$

$$A = \begin{pmatrix} 1/4 & 1/4 - \sqrt{3}/6 \\ 1/4 + \sqrt{3}/6 & 1/4 \end{pmatrix} \approx \begin{pmatrix} 0.25 & -0.039 \\ 0.539 & 0.25 \end{pmatrix}.$$

$A_{12} = -0.0387$ sits above the diagonal, so stage one depends on stage two. There is no order in which these can be evaluated one at a time.

The closed forms at $q = 1, 2, 3$ are hard-coded and used to check the general construction from the Legendre roots, which is what builds every tableau actually used.

Picture. Explicit stepping walks across the room with your eyes on your feet. Collocation draws one smooth curve to the far wall and then checks, at q chosen points, that it is heading where the physics says.

The discrete-time construction: read the scheme backwards and the labels disappear

The network is asked to output all stage states and the endpoint at once,

$$\mathcal{N}_\theta : S_0 \longmapsto (S_1, S_2, \dots, S_q, S_{\text{end}}).$$

Each of the $q + 1$ scheme equations is then rearranged to express the *starting* state in terms of a proposed output:

$$S_0^{(j)} = S_j - \Delta z \sum_k A_{jk} f(S_k, z_0 + c_k \Delta z), \quad S_0^{(\text{end})} = S_{\text{end}} - \Delta z \sum_k b_k f(S_k, z_0 + c_k \Delta z).$$

Each output produces its own **reconstruction** of the starting state, which is known because it is the input. The discrepancies are the entire training signal:

$$r_{n,d}^{(j)} = \frac{S_{0,n,d}^{(j)} - S_{0,n,d}}{\sigma_d}, \quad \mathcal{L}_{\text{phys}}(\theta) = \frac{1}{(q+1) d_{\text{max}} N} \sum_{n,j,d} \left(r_{n,d}^{(j)} \right)^2, \quad d_{\text{max}} = 4.$$

Nothing in that loss is a label. What enters is the starting state, which is the input; the tableau, which is arithmetic; and the field, which is a measured map. The field enters through f evaluated at the positions the network itself proposes, so it must be available in a differentiable form.

Picture. Hand someone a proposed route and ask them to walk it backwards from every waypoint. If all $q + 1$ backwards walks land on the entrance, the route solves the scheme. Nobody was ever told the correct route.

The scales in the loss are measured from the population, and they are label free

The components differ by decades: x is hundreds of millimetres, t_x a few tenths and dimensionless. Squaring and summing them directly would let x own the loss. Each residual component is divided by a fixed scale σ_d before squaring, which also makes the loss dimensionless.

Measured on the full training split, all leg types

$$\sigma_x = 655.6 \text{ mm}, \quad \sigma_y = 360.2 \text{ mm},$$

$$\sigma_{t_x} = 0.2592, \quad \sigma_{t_y} = 0.1152, \quad \sigma_{\text{qop}} = 0.1190.$$

On the frozen cross-magnet leg: $\sigma_x = 176.5 \text{ mm}$ and $\sigma_y = 182.7 \text{ mm}$ on the input side, with wider output-side values of 396 mm and 417 mm reflecting the spread the states acquire after crossing the magnet.

One scale per component for the whole population is a deliberate choice, inherited unchanged from the verified July baseline. On a single frozen leg it is unobjectionable, because the population is one geometry.

On a population spanning steps from 70 mm to 8.6 m it is not. It turns out to be the leading suspect for the general network's failure on short steps, and it is what the residual parameterisation three slides from now replaces.

The scales are computed from the spread of each component over the training states themselves. No reference trajectory is consulted, so the normalisation does not smuggle a label into a label-free loss. Every experiment recomputes them on its own dataset.

The straight-line-residual parameterisation, and the bending integral that scales it

Exactly one thing changes: what the network is asked to output.

$$\text{output}_j = \text{straight}_j + \text{scale} \odot \text{net}_j, \quad \text{straight}_j = (x_0 + t_x(z_j - z_0), y_0 + t_y(z_j - z_0), t_x, t_y),$$

so net_j is a pure number and the network predicts the magnet's whole contribution and nothing else.

The scale may not use a label, so it is built from the inputs

$$|\Delta t_x| \approx \kappa |\text{qop}| l_B, \quad |\Delta x| \approx \frac{1}{2} \kappa |\text{qop}| l_B |\Delta z|,$$

$$l_B = \int_{z_0}^{z_0 + \Delta z} |\mathbf{B}(x_s(z), y_s(z), z)| dz \quad [\text{T mm}],$$

the field magnitude integrated along the *straight line* x_s, y_s , by a 16-point midpoint rule. Every symbol on the right is an input. Floors of 10^{-3} mm and 10^{-6} keep it away from zero.

Why this makes every leg order one

- A 70 mm sensor hop and a 5.2 m magnet crossing differ by five orders of magnitude in the physical correction, and by the same five orders in $l_B |\Delta z|$. The ratio the network emits is order one in both.
- The estimate does not have to be right. It has to be right to within a factor of a few on every leg type.
- **The gate, fixed before looking:** take a flat node profile unless its endpoint medians fall outside 0.1 to 10 on some leg while a polynomial profile's do not. Medians came out 1.09, 1.03, 0.95 on the three leg types, so flat was selected.
- It is zero in a field-free region, so **the trivial baseline is inside the parameterisation** and the network cannot be worse than a straight line unless it is actively wrong.

The continuous-time objective, written out, and what it does not constrain

For the van der Pol system

$$\frac{dy_1}{dt} = y_2, \quad \frac{dy_2}{dt} = \mu(1 - y_1^2)y_2 - y_1, \quad \mu = 1,$$

the network $y_\theta(t)$ is trained on

$$\mathcal{L} = \underbrace{\frac{1}{N} \sum_{i=1}^N \|\dot{y}_\theta(t_i) - f(y_\theta(t_i))\|^2}_{\text{residual, over collocation points}} + \underbrace{\|y_\theta(0) - y_0\|^2}_{\text{one anchor}},$$

with $\dot{y}_\theta$ from automatic differentiation.

The whole of the physics is a *mean* over N points. The whole of the boundary information is a *single* number. That asymmetry is the mechanism of everything that follows.

The system's one fixed point, and its Jacobian

$$J(y) = \begin{pmatrix} 0 & 1 \\ -1 - 2y_1y_2 & 1 - y_1^2 \end{pmatrix}, \quad J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix},$$

eigenvalues $\lambda = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$. Both real parts are $+\frac{1}{2}$: the origin is an **unstable spiral**, not a saddle. Displacements grow like $e^{t/2}$ while rotating.

And yet $f(0,0) = 0$. So a constant output at the origin has $\dot{y}_\theta = 0$ and $f(y_\theta) = 0$, hence **residual exactly zero at every collocation point, wherever those points sit**. The residual term takes its global minimum there. Only the anchor prefers the truth.

The fixed point is only the cheapest member of a family, and the family gets cheaper with T

The splice construction

Fix the collocation set and a time t_0 . Let $\alpha(t)$ be smooth, equal to 1 near every collocation point before t_0 , equal to 0 near every collocation point after it, and 0 for all $t \geq t_0$. Then $y^\dagger = \alpha y^*$ satisfies the anchor and has residual exactly zero at every collocation point: near each point it coincides either with the true solution or with the zero function, and both solve the equation. The residual is nonzero only inside the transition layer, which has been placed *between* points.

For an autonomous system any trajectory may be spliced in, not only the zero function. Two other members appear in my results: the phase-shifted loop, because the residual cannot see phase, and off-loop trajectories entering from far outside the cycle.

The impostors are not approximate solutions. They are exact ones, joined to the truth where the loss cannot look.

What resampling changes, and the $1/T$ price

If fresh points are drawn every step the layer is eventually sampled. Inside a layer of width h the residual is of order $|y^*(t_0)|/h$, so

$$\frac{1}{T} \int_{\text{layer}} \left| \frac{y^*(t_0)}{h} \right|^2 dt \sim \frac{|y^*(t_0)|^2}{h T}.$$

A hand-over from the loop to the origin over about a period ($h \approx 2$) at amplitude 2 in a window $T = 27$ costs of order $4/(2 \times 27) \approx 0.07$.

Three effects push the same way as T grows: the price of a splice **falls like $1/T$** ; the number of places to put one grows like T ; and the price of the truth rises, because the network must hold $T/6.663$ oscillations in phase against a single anchor. **No resource changes the price of a splice.** That is why size, density and placement all came back null.

Pseudo-time stepping is the only fix whose mechanism prices the whole family

The residual at training step k is replaced by a relaxed version that also penalises change from the previous iterate:

$$L_k(\theta) = \frac{1}{N} \sum_{i=1}^N \left\| \frac{y_\theta(t_i) - y_{\theta_{k-1}}(t_i)}{\tau_k} + r_\theta(t_i) \right\|^2 + L_{\text{IC}}(\theta),$$

with fresh collocation points every step. This is the implicit Euler discretisation of the artificial dynamics $\partial u / \partial s = -r[u]$ in a pseudo-time s .

One pseudo-time update of a spliced candidate turns its hidden layer residual of order h^{-1} into order $h^{-1} + \tau^2 h^{-3}$, an amplification by τ^2/h^2 . The relaxed objective therefore penalises spliced candidates far more than the plain residual does, and it does so for *every* member of the family rather than for one.

Implementation notes

- τ is set adaptively from a finite-difference surrogate of the residual Jacobian. I follow the authors' reference code, which produced their published numbers, where it and their text disagree on the late-training shrink of τ .
- In my runs the adaptive step settled between 0.5 and 0.9 in every case.

Predictions I recorded before the runs

- The regulariser closes exactly one member of the family and converts parking into some other member at long horizons.
- Resampling alone produces a parked stationary point at the layer scale, of order 10^{-2} .
- Causal weighting delays the splice without removing its incentive.
- Longer training changes nothing, because an exact minimiser cannot be trained away, only revealed.

Three fixed comparators and one metric definition, used everywhere

The comparators

1. **The exact scheme.** The same q -stage Gauss–Legendre equations, on the same states, solved by a root-finder in double precision, no network anywhere. Convergence declared below 10^{-8} in millimetres and milliradians. This is the *ceiling*: the best any network trained on those equations could possibly do.
2. **The data twin.** Identical network, identical initialisation, identical optimiser, identical budget, physics loss swapped for mean squared error against reference stage and endpoint states. The supervised control, and the same control the source work used.
3. **The straight line.** Ignore the magnet, continue along the incoming slopes. Any surrogate that does not beat it is doing nothing useful.

The twin is drawn next to the physics network in every result, and ceilings are always quoted on the population being scored.

The metrics

- **Endpoint error:** the larger of $|\Delta x|$ and $|\Delta y|$ against the double-precision fourth-order Runge–Kutta reference at the end plane, in micrometres. Median over test states, with the 95th percentile where the tail matters.
- **Stage error:** the same, over all q interior stage states at once.
- **Capped relative error**, shared with the van der Pol study:

$$\rho = \min\left(1, \frac{\|S_{\text{ref}} - S_{\text{pred}}\|^2}{\|S_{\text{ref}}\|^2}\right).$$

The cap exists because the denominator can approach zero, and a miss larger than the state carries no more information than “completely wrong”.

- **Relative L_2** on van der Pol, over all evaluation times and both components.

Over seeds: the median of the per-seed medians, range in brackets. **A single seed is not a measurement.**

Causal weighting: it moves the failure by one period and makes the rest visible

Wang, Sankaran and Perdikaris (2022)

What it proves. A plain residual objective fits the whole window at once, so it can satisfy late times before early times are correct. Weighting each collocation point by how well the trajectory behind it already satisfies the equation makes a front of large weights sweep outward from the anchor, restoring temporal causality to the optimisation.

Its protocol. Weights w_i from a running exponential of the cumulative residual, with a tolerance ε annealed over training.

My implementation check. Weight profiles logged through training; the front's position and the minimum weight are stored per run, so a run that ends mid-domain declares itself.

My result. It moves the cliff from one period to two, and no further.

periods	plain	causal	front progress
0.45	3.8×10^{-5}	4.8×10^{-5}	met, 3–5.5k
1.05	6.6×10^{-4}	7.6×10^{-5}	met, 8.5–28.5k
2.10	0.79	2.1×10^{-4}	epoch limit
4.05	0.90	1.90	limit, mid-domain
6.00	0.94	2.04	limit, mid-domain

Cost of crossing the domain grows faster than linearly: about 3.5k Adam epochs at half a period, 10 to 30k at one, 60k at two, beyond the limit at four. Runs stopped mid-domain score *worse* than plain, because the far part of the domain still has near-zero weights and is unconstrained.

The budget-extension test settles it: the seven runs that hit the 60,000-epoch limit were rerun at 240,000, four times the budget, with nothing else changed. **None of the seven completed the crossing.**

Fixed points as global minima: the diagnosis, with no remedy offered

Rohrhofer, Posch, Gößnitzer and Geiger (2023)

What it proves. If the network output is a constant equal to a fixed point y^* with $f(y^*) = 0$, the residual is exactly zero at every collocation point wherever they sit. Fixed points are therefore global minima of the residual term, with basins that grow with the horizon, and only the initial-condition term prefers the true solution.

What they measure. Success falls monotonically with the horizon and with the proximity of the start to a fixed point. Larger networks repair nothing, so they conclude the difficulty is bound to optimisation complexity rather than expressive power. Near a fixed point the spurious minimum can be *better* valued than the true one.

They propose no remedy, and do not distinguish the fixed point from other zero-residual candidates.

My implementation check. The failure classifier reads the trained residual on the reference grid and calls a run *parked* if more than 30 percent of the window sits within 0.2 of the origin, so the diagnosis is measured per run rather than asserted from a plot.

My result. Confirmed exactly, and then superseded as a complete account.

- The plain objective parks in 9 of 10 runs at four periods and 10 of 10 at six.
- The parked fits on a fixed collocation set reach **the same loss as a success**, which is the silent-failure mode in its converged form.
- But closing the origin alone does not fix the problem, because **the origin is not the only zero-residual candidate**. That is the next two slides.

The stillness regulariser closes the origin, and the optimiser takes the next member

Babic, Rohrhofer and Geiger (2025)

What it proposes.

$$C \times \frac{1}{N} \sum_i R_{\text{SE}}(t_i) R_{\text{LS}}(t_i),$$

$$R_{\text{LS}} = \sum_{\lambda} \max(\operatorname{Re} \lambda, 0), \quad R_{\text{SE}} = \exp(-\|\dot{y}_{\theta}\|^2/\varepsilon),$$

the eigenvalues being those of the Jacobian at the network's own output, $\varepsilon = 0.01$, and C decaying linearly from 0.5 to zero at half the training. It penalises a candidate for *sitting still inside a locally unstable region*.

On the true loop the slowest speed squared is 0.59, so the term is inert on the correct solution by construction. Their van der Pol demonstration stops at $T = 15$, about two periods, and their published protocol resamples 1024 points every epoch, so the published “regularised” method is **the regulariser plus resampling**.

My implementation check. Implemented as published; the eigenvalue term checked against a numerical eigen-decomposition, and its gradients confirmed finite. Run in three variants: published dose, held at 0.5 throughout and kept in the polish, and the published regulariser plus resampling protocol.

My result. **It works exactly as far as they tested it**, and converts the failure rather than removing it.

- Two periods: 10 of 10, up from the plain objective's 5.
- Four and six periods: 0 of 10 in all three variants.
- The failures are no longer parks. Every variant converges onto a **flow-following splice**: residual below 0.05 on more than 95 percent of the window, one spike of 10^2 to 10^3 immediately after $t = 0$, entering from $y_1 \approx 4.5$ and riding onto the loop out of phase, at a converged loss of 10^{-5} to 10^{-4} .
- Keeping the term on, or adding resampling, changes nothing. As predicted.

Pseudo-time stepping prices the whole family, and it is the only thing that works past two periods

Wang, Koohy, Lu and Perdikaris (2026)

What it proves. Their Theorem 2.1 is the splice construction of slide 13: for any fixed collocation set there exists a candidate with *exactly* zero empirical residual that is not the solution. The proof uses nothing about the origin except that it solves the equation. They note that under resampling a layer of width h still costs only order h^{-1} , and propose the relaxed objective of slide 14.

My implementation check. Their reference code's adaptive τ ; fresh collocation points every step; the same plateau-then-polish convergence rule as every other arm, so no arm gets a budget advantage.

τ settled between 0.5 and 0.9 in every run, which is a useful sanity check that the adaptive rule is not degenerating.

My result. The only arm that survives past two periods, and its own wall is measured.

- Two periods 10/10, four periods 8/10, six periods 7/10, where every other arm is 0/10 past two.
- The eight successes at four periods span 8.5×10^{-5} to 8.4×10^{-4} ; the seven at six periods span 4.2×10^{-4} to 8.9×10^{-3} .
- **Its own wall lies between six and ten periods:** 1/10 at ten (a marginal success at 5×10^{-2} with the polish at its cap) and 0/10 at fifteen.
- The failure mode does not change with the horizon, it arrives later: a late park after four to eight good periods.

And the $1/T$ price is confirmed quantitatively: over the 65 parked runs whose transition layer is actually sampled, $\text{loss} \times T$ has median 0.85 and range 0.84 to 0.90 from four to fifteen periods.

Van der Pol first, because there the truth is known and the failures are the method's own

$$\frac{dy_1}{dt} = y_2, \quad \frac{dy_2}{dt} = \mu(1 - y_1^2)y_2 - y_1, \quad \mu = 1.$$

For $|y_1| < 1$ the damping term pumps energy in; for $|y_1| > 1$ it takes energy out. The result is a limit cycle onto which every trajectory except the equilibrium spirals.

- Period [6.663287](#), loop extent $|y_1| \leq 2.009$, $|y_2| \leq 2.678$, both measured from the reference.
- The canonical start is $(2, 0)$, a point on the loop.

Why start here

- **Known truth.** A reference trustworthy to about 10^{-13} , six orders below anything a network reaches, so the yardstick never contaminates the measurement.
- **Cheap.** Sweeps of hundreds of runs are affordable, so resource axes can be measured to exhaustion rather than argued about. This talk reports over a thousand van der Pol runs.
- **Isolating.** Any failure is the method's own. No detector, no field map, no simulation sample and no label pipeline to blame it on.

Every design choice in the LHCb work that could have been argued about was settled here first: stage count, the optimisation-floor diagnosis, the seed-selection protocol, the supervised control, and chaining a one-step map rather than taking one giant step.

The same machinery on both problems

- **A verified reference integrator, not a trusted one.** On van der Pol, a seven-stage sixth-order explicit Runge–Kutta method at step 10^{-3} , verified four ways. On LHCb, fourth-order Runge–Kutta in double precision at 5 mm steps, checked by four integrity gates on the labels it produces.
- **A Gauss–Legendre tableau for any q ,** built from the Legendre roots, checked against the closed forms at $q = 1, 2, 3$ and against the exactly solved scheme.
- **One-step network, both losses.** The same shape, the same optimiser, the same budget. Only the loss changes.
- **A data-trained twin as the control** in every experiment, so every claim about the label-free loss has a matched supervised number beside it.

The convergence-assured protocol

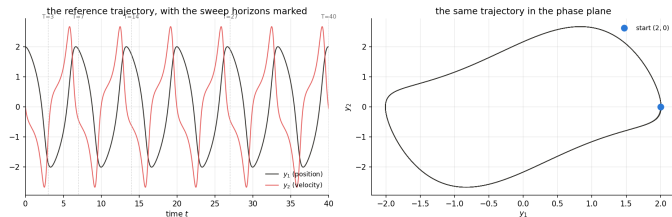
1. Train with full-batch L-BFGS (limited-memory Broyden–Fletcher–Goldfarb–Shanno), 200 iterations per restart, strong Wolfe line search, history 120.
2. **Stall:** restart until two consecutive restarts each improve the training loss by less than one percent.
3. **Confirm:** start a completely fresh optimiser with its curvature history discarded. The run counts as converged only if it re-stalls within two restarts without moving the train, validation or test medians by more than one percent.
4. Unconverged runs are **never** pooled into a median.

This exists because of a July lesson: a first attempt used a fixed budget of six restarts, and every run was still improving by 20 to 35 percent per restart when the budget ran out. That produced a factor-two gap between the two losses which turned out not to be real. Single thread, double precision, one job per architecture, mode and seed on the Nikhef farm.

The dictionary from van der Pol to LHCb

van der Pol	LHCb	what changes
independent variable t	the beam coordinate z	nothing formal
state (y_1, y_2) , two components	state (x, y, t_x, t_y, qop) , four predicted	dimension, and units that differ by decades
parameter μ , fixed at 1	charge over momentum, carried through unchanged	it is now an <i>input</i> , and it is conserved
autonomous: the right-hand side sees only the state	non-autonomous: the right-hand side sees z through the field	the hard structural difference
analytic right-hand side	a measured table, trilinear, C^0 and not C^1	the ceiling of the whole exercise
a training rectangle of starting states	real event states harvested from simulation	the population is given, not chosen
order-6 reference at step 10^{-3} , error $\sim 10^{-13}$	fourth-order Runge–Kutta at 5 mm	the reference is 10^{-13} there, about $20 \mu\text{m}$ here
difficulty axis: the number of periods in the window	difficulty axes: the bending integral I_B of a leg, and the number of chained legs	two axes instead of one

Van der Pol data: eight starting points against one reference verified four ways



Left: the reference from (2, 0) over six periods, with the five training horizons of the continuous-time experiments marked. Right: the same trajectory in the phase plane, the closed loop to be learned.

Verified, not asserted

1. Convergence order, as the slope of global error against step size on three problems with closed-form solutions: measured 6.06, 6.27, 5.75 against the theoretical 6.
2. Step-halving self-consistency reaches a rounding floor of 6.4×10^{-14} below $h \approx 5 \times 10^{-3}$, so at the working step the truncation error is beneath double-precision rounding.
3. Period 6.663287 against the accepted 6.6633.
4. Closure of the limit cycle onto itself after one full period: 5.6×10^{-14} .

Eight starting points integrated to $t = 40$. Used only for scoring, never for training, with the single deliberate exception of the data-trained twin. Where a network's evaluation times fall between stored samples the integrator is re-run to stop exactly there; nothing is interpolated.

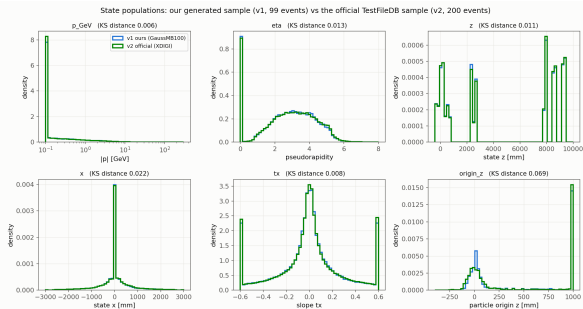
LHCb data: 323,533 legs of four kinds, harvested from an official sample

leg	what it is	typical step	rows
A	vertex fetch	341 mm, backward	49,014
B	cross-magnet	5175 mm, both ways	41,398
C	plane-to-plane	70 mm, forward	205,887
D	downstream to vertex	−8.6 m	27,234
total			323,533
split by particle, seed 20260718			258,857/32,389/32,287

The harvest: dump the simulation truth; turn each tracker hit into a state with position, local slopes and truth momentum; form legs from each particle's own geometry; label by fourth-order Runge–Kutta at 5 mm; split by particle, because legs from one particle are highly correlated and splitting by leg would leak.

Type D is built and labelled but **never trained on**: it is an out-of-training test by construction.

Official simulation, 2024 minimum-bias, 200 events from five of eight files mirrored locally. A deliberate policy change on 21 July: small configuration errors in home-grown generation cause large and silent problems.



The harvested population against the in-house sample it replaced: momentum, pseudorapidity, the z of the harvested states, transverse position, slope, and the z of the production point, with Kolmogorov–Smirnov distances printed. Distances 0.006 to 0.069, so switching sample changed the provenance of the population and not its physics. The three clusters in the z panel are the detector itself.

Four gates on the labels, and the one percent of states that carried half the loss

gate	what it checks	result
G1	closure: propagate forwards, then backwards	median 1.4 nm
G2	truth: against the particle's own next hit	median 22 μm
	the same, above 5 GeV	6.3 μm
	the same, below 2 GeV	86 μm
G3	step: repeat the integration at 1 mm, not 5 mm	median 8.9 nm
G4	conservation: qop passes through unchanged	exact

G2 defines the scope of the project. The label disagrees with the real next hit by an amount growing as $1/p$, the signature of multiple scattering. That residual is not an error in the labels; it is the material effect, deliberately outside the surrogate's remit.

The fiducial requirement, and why it is the portability lesson

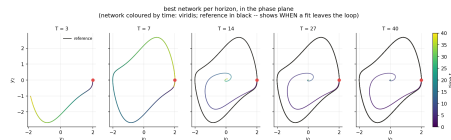
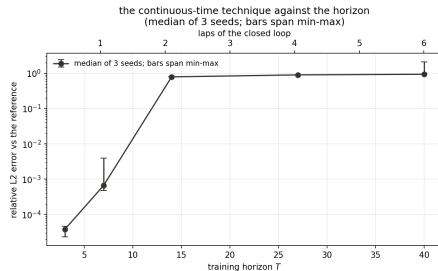
- First run against the official sample: the physics loss appeared far worse than the twin, 403 to 595 μm against 175 to 213 μm .
- Cause: **21 of 2000 training states**, one percent, are soft tracks of 1.2 to 1.5 GeV that bend so hard they leave the field map, past $|x| = 4$ m, outside the LHCb acceptance as well. Those 21 states carried **half of the entire physics loss**; the top one percent carried 68 to 78 percent.
- **The asymmetry is structural.** The physics loss evaluates the field where the network proposes, so a trajectory outside the map is asked to satisfy equations built on a clamped, meaningless field. The twin never touches the field: it fits labels computed with the same clamped field and absorbs the pathology silently.
- With the requirement in place, runs stopped stalling at a plateau, taking 147 to 152 restarts instead of 57 to 72, and the training loss fell from about 1.1×10^{-4} to 8.3×10^{-6} , the same range as the twin.

The continuous-time construction is excellent below one period and collapses beyond it

T	periods	N	rel. L_2 (median)	final loss
3	0.45	60	3.8×10^{-5}	$1.4\text{--}2.0 \times 10^{-7}$
7	1.05	140	6.6×10^{-4}	$2.3 \times 10^{-6}\text{--}2.8 \times 10^{-5}$
14	2.10	280	0.79	$\approx 6 \times 10^{-2}$
27	4.05	540	0.90	$\approx 3 \times 10^{-2}$
40	6.00	800	0.94	$2.1\text{--}4.8 \times 10^{-2}$

Collocation density fixed at 20 points per unit time; medians over three seeds. Best and worst seeds at $T = 14$ are 0.789 and 0.805: **the seeds agree to two decimal places**, so the failure is systematic and not an initialisation accident.

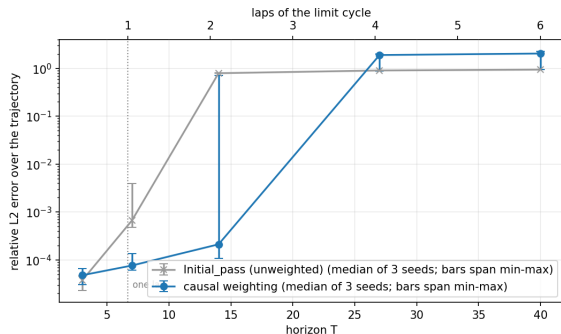
Below one period this is at or better than the accuracy the source work reports for its own continuous-time demonstrations, 1.97×10^{-3} and 6.7×10^{-4} . Unsurprising for a two-state oscillator against partial differential equations, but it confirms the machinery is correctly built.



Top:

relative L_2 against training horizon, median of three seeds with min to max whiskers. Bottom: the failed fits in the phase plane, best seed per horizon. Past one period the fit abandons the closed loop and spirals into the origin, *against* the direction of the true flow.

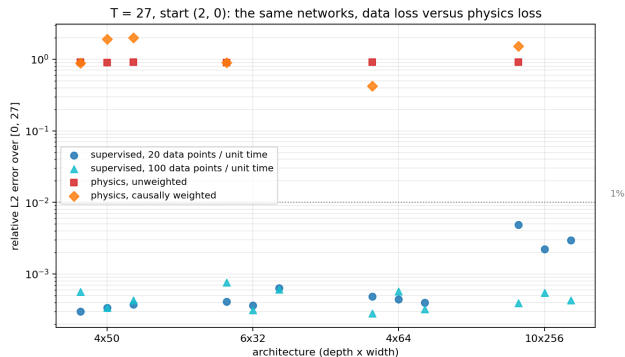
Causal weighting moves the cliff to two periods and turns silent failures into visible ones



Same metric, same reference, same collocation density on both sides; the causally weighted runs add an Adam phase of up to 60,000 epochs before an identical final L-BFGS phase. Median of three seeds with min to max whiskers.

- At two periods, where the plain objective fails at 0.79, causal weighting reaches 2.1×10^{-4} : three to four orders of magnitude. In the capped measure the contrast is about seven orders, mean ρ of 6.2×10^{-1} against 4.0×10^{-8} .
- **The essential difference is visibility.** These failures declare themselves: the minimum weight and the front position show it still mid-domain. The plain failures reported themselves as converged.
- One run at the epoch limit produced a second kind of silent failure, and it is the one that matters: **training loss 9.7×10^{-6} with error 0.71**, an out-of-phase near-solution joined to the truth by a jump inside an unsampled interval, invisible both to the loss and to weights computed on the same grid.
- That single run is the empirical form of the splice theorem, found before I knew what it was.

The same networks fit four periods from data, so the objective fails and not the network



Every supervised run at four periods, at both sample densities, against both physics variants, by architecture. The separation is three to four orders of magnitude with the network, the sample times and the optimiser caps held fixed.

The design

- Same networks, same sample-time distribution, same optimiser, same caps, at four periods. **Only the loss changed:** plain mean squared error to the known states, no residual anywhere.
- Four architectures (4×50 , 6×32 , 4×64 , 10×256), two sample densities (20 and 100 per unit time), three seeds: **24 runs**.

The result

- All 24 fit four periods**, at relative L_2 between 3×10^{-4} and 5×10^{-3} .
- Every physics-trained run at the same horizon sits between **0.4 and 2.0**.

So the network can hold four periods in phase against a single anchor and the optimiser can find those weights. What it cannot do is find them *through the residual*. A supervised success shows only that the network *could* express the answer; it does not say which change to the objective would arrive there.

Eight objectives trained to convergence: only pseudo-time stepping survives past two periods

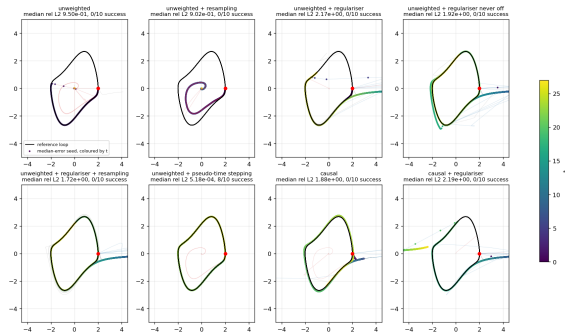
arm	$T = 14$	$T = 27$	$T = 40$	median rel. L_2 , 27 / 40	failures past two periods
plain	5/10	0/10	0/10	0.95 / 0.94	parked 9, diffuse 1 (10 parked)
resample	10/10	0/10	0/10	0.90 / 0.94	parked 10 at loss 0.02–0.03 (10)
regulariser	10/10	0/10	0/10	2.2 / 3.6	flow-following 6, diffuse 3, parked 1 (7, 3)
regulariser always on	10/10	0/10	0/10	1.9 / 2.7	flow-following 8, diffuse 2 (5, 5)
regulariser + resampling	10/10	0/10	0/10	1.7 / 2.6	flow-following 6, diffuse 4 (5, 5)
pseudo-time stepping	10/10	8/10	7/10	5.2×10^{-4} / 5.4×10^{-3}	parked 2 (parked 2, diffuse 1)
causal	8/10	0/10	0/10	1.9 / 2.1	parked 2, flow-following 4, diffuse 4
causal + regulariser	7/10	0/10	0/10	2.2 / 2.1	flow-following 7, diffuse 2, parked 1

Protocol. All arms share a 4×50 network, the start $(2, 0)$, density 20, seeds 0 to 9, at $T \in \{14, 27, 40\}$ (2.1, 4.1 and 6.0 periods). A fixed-budget first pass of 150 runs cured $T = 14$ but left the regularised runs still descending when cut off, so **every run reported here was trained to a demonstrated plateau**: Adam at 10^{-3} until the objective improved by less than 10^{-4} over the trailing 10,000 epochs, cap 200,000, then an L-BFGS polish of up to 60 restarts with the stall rule. 239 of 240 satisfy the rule.

Reading it. Success is relative $L_2 < 0.15$, the threshold of the two papers proposing the fixes. **Every fix works at two periods**, which is exactly where the regulariser paper tested it: the plain objective's 5 of 10 becomes 10 of 10 under resampling, under the regulariser, or under pseudo-time stepping. Past two periods every arm is 0 of 10 except one. Converged and budget-limited runs are never pooled, and every non-success is classified from its residual rather than by eye.

The three failure classes, and why loss stops predicting error on one of them

T = 27: phase portraits (all seeds faint, coloured by outcome; the median-error seed time-coloured)



- **Parked** (plain, resampled). Follows the reference for about one period, then sits at the origin. On a fixed set the parks reach the *same loss as a success*. Under resampling they sit at loss 0.02 to 0.03 with zero gradient, which is exactly the stationary point at the layer scale that the $1/T$ formula predicts.
- **Flow-following** (every regulariser variant). Satisfies the anchor, splices within the first unsampled gap onto a trajectory entering from $y_1 \approx 4.5$, and rides it onto the loop out of phase, at a converged loss of 10^{-5} to 10^{-4} .
- **Diffuse**, otherwise.

Two facts from the loss-term curves. In the regularised arms the raw regulariser value *rises* after switch-off while the objective keeps falling. And the polish drives the flow-following objectives below 10^{-4} **without moving the error at all**. Loss and error decouple on that branch. That is the silent failure of the first result slide, in its fully converged form.

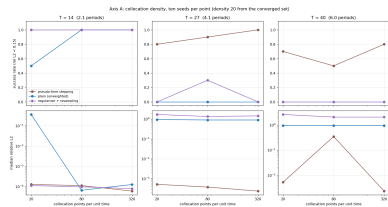
The two resource axes: density buys seeds at four periods and nothing at six

arm	collocation density, $T = 27 / T = 40$			horizon T , at density 20				
	20	80	320	14	27	40	67	100
pseudo-time stepping	8 / 7	9 / 5	10 / 8	10	8	7	1	0
plain	0 / 0	0 / 0	0 / 0	5	0	0	0	0
regulariser + resampling	0 / 0	3 / 0	0 / 0	10	0	0	0	0

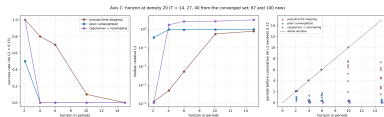
Successes out of ten. 240 runs, ten seeds per cell, same protocol as the eight-arm table. Horizons 67 and 100 are ten and fifteen periods.

- At two periods, density 80 lifts the plain objective from 5 of 10 to 10 of 10, so **an earlier null verdict on density, obtained at fixed budget with L-BFGS alone, does not survive training to convergence**. At four periods and beyond it still fails at every density, so that verdict stands.
- The published regulariser protocol has its first successes past two periods at density 80: 3 of 10 at four periods, one converged and two budget-limited. At density 320 it is back to 0 of 10.
- Pseudo-time stepping's parking rate at four periods falls with density, 8, 9, 10. At six periods it does not move in any direction, 7, 5, 8, with different seeds parking at each density and every parked run at the same loss whatever the density. **Ten seeds cannot separate the six-period rates**.
- Cost grows roughly linearly with the number of points: the density-320 runs at $T = 40$, with 12,800 points, took about four hours each.

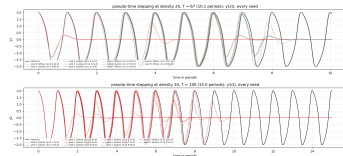
The two axes as pictures: a rate that moves with density, and a wall that does not



Success rate (top) and median relative L_2 (bottom) against collocation density, one column per horizon, ten seeds per point.



Success, median relative L_2 , and the number of periods survived before the cumulative relative L_2 exceeds 0.15, per run, against the horizon in periods.

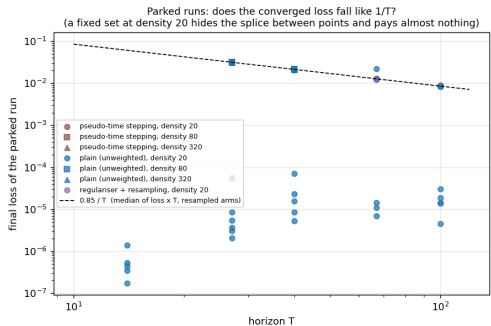


Pseudo-time stepping at ten and fifteen periods, y_1 against time in periods for every seed, coloured by class.

The failures at long horizon are late parks: the network rides the loop for four to eight periods, drifts off, and sits at the origin. **Nothing wanders off the loop.** The failure mode does not change with the horizon, it arrives later.

One consequence for reporting: the classifier reads a late park as diffuse, so the periods-survived panel is the right readout for long windows, not the success rate.

The $1/T$ price is not a story, it is a measured line



converged loss of every parked run against T , with the $1/T$ line drawn through the median of loss $\times T$ over the resampled arms.

- Over the 65 parked runs whose transition layer is actually sampled (the resampled arms at any density, and the plain arm at densities 80 and 320), **loss $\times T$ has median 0.85 and ranges 0.84 to 0.90**, from $T = 27$ to $T = 100$.
- The plain arm's parked runs at density 20 sit three to four orders of magnitude lower, because a fixed set of 20 points per unit time leaves room to hide the layer. At density 80 the gap is too small to hide in, and a fixed set behaves like resampling.
- Two consequences.** The $1/T$ prediction holds quantitatively. And **more density cannot move a resampled method's wall by pricing**, because the parked branch is already fully sampled, so denser points do not raise its price. What density can do is make the descent to the truth cheaper than the descent to the park in more seeds, which is what the four-period column shows and the six-period column does not.

The wall between six and ten periods is a *late park*: the network rides the loop for four to eight periods, drifts off, and sits at the origin. Nothing wanders off the loop. What sets its position is not yet measured.

The discrete-time column has no such failure, and its floor is not the scheme

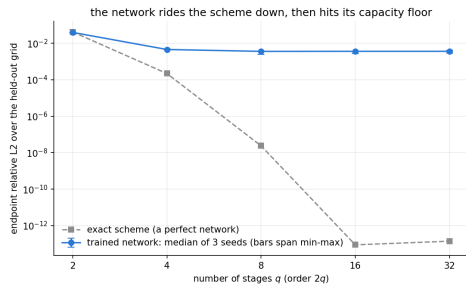
q (order $2q$)	trained network	exact scheme
2 (4)	3.94×10^{-2}	4.20×10^{-2}
4 (8)	4.57×10^{-3}	2.20×10^{-4}
8 (16)	3.61×10^{-3}	2.38×10^{-8}
16 (32)	3.63×10^{-3}	8.9×10^{-14}
32 (64)	3.63×10^{-3}	1.4×10^{-13}

Endpoint relative L_2 on the 441 held-out states, median over three seeds, against the exactly solved scheme on the same states and metric. Step $\Delta t = 0.8$.

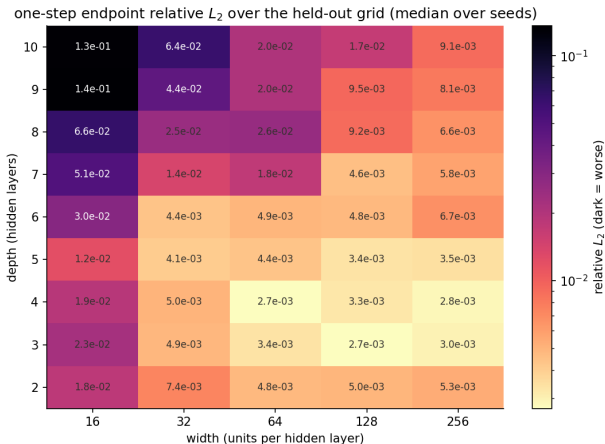
Capped measure, same runs: median ρ falls 7.2×10^{-5} , 1.0×10^{-5} , 6.5×10^{-6} , 6.0×10^{-6} , 5.1×10^{-6} . And $\sqrt{6.5 \times 10^{-6}} \approx 2.5 \times 10^{-3}$, consistent with the relative L_2 floor as it must be.

- All fifteen runs converge to losses of 7×10^{-6} to 2.6×10^{-5} , with no collapse and no divergence. And, unlike every failed continuous-time run, **the loss is an honest proxy for the error**: a squared-residual loss of 10^{-5} resolves outputs to roughly its square root, which is where the held-out error lands.
- At $q = 2$ the network sits *on* the exactly solved scheme's own error: the map is scheme-limited, and even a perfect fit would be that wrong.
- From $q = 4$ the curves part. The scheme drops ten orders of magnitude to roundoff; the network stays flat at **3.6 to 4.6×10^{-3}** , within a factor 1.27 for all $q \geq 4$, all seeds agreeing.
- Asking for 66 outputs at $q = 32$ instead of 18 at $q = 8$ degrades nothing, and the largest errors sit at the corners of the training region where the dynamics are fastest.
- So all later work fixes $q = 8$: the scheme's own error is five orders below the network's, and nothing about the network is masked by the scheme.

The same floor twice: against stages, and across forty-five architectures

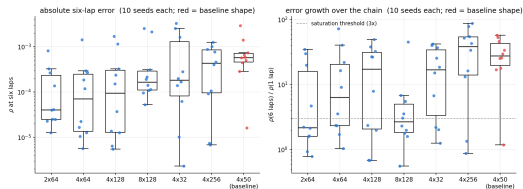


Endpoint relative L_2 against the number of stages: the trained network (median of three seeds, min to max whiskers) rides the scheme's error down and then flattens onto its own floor, while the exactly solved scheme continues to machine precision.



One-step error over the architecture grid at fixed $q = 8$: depth $\times$ width, three seeds each. No entry approaches the exactly solved scheme; depth harms beyond about five layers, width stops helping beyond 64 to 128 units, and the worst corner sits at 1.3×10^{-1} . Two pictures, one statement. **More stages cannot help** once the scheme's error is below the network's, and **more parameters cannot help either**, because the binding constraint in both cases is how far the optimiser got. That is the finding

On van der Pol the floor is the optimiser, and the horizon is decided by the seed

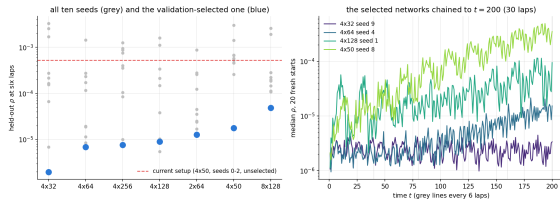


Ten seeds for each of seven architectures, six-period chained error. The seed-to-seed spread within one architecture rivals the whole grid's spread.

Design. Depth $\{2, \dots, 10\}$ crossed with width $\{16, 32, 64, 128, 256\}$, 45 architectures at $q = 8$, three seeds, 135 runs, everything else frozen. Seven architectures then at ten seeds. The 108 test starts are split: 50 used only to rank, 58 held out and used only to report.

- **The floor does not move.** Best of 45 is 2.5×10^{-3} , a factor 1.5 below the 4×50 configuration, and still five orders above the exact scheme. Depth beyond about 5 hurts sharply; width stops helping beyond 64 to 128.
- **The binding constraint is the optimiser.** Spearman rank correlation between converged training loss and held-out one-step error over the 135-run grid: $+0.974$. Architectures that fail also fail to train. No capacity-rich, badly-generalising regime exists.
- **But that same loss ranks six-period chained error at only $+0.35$.** Long horizons are decided by the seed: the growth ratio varies $13\times$ between seeds of one architecture and $300\times$ across the grid.

Growth is a measurable, transferable property, so select the seed on the chain



The selection payoff and the far-horizon check: validation-selected networks against their unselected siblings, and the selected network holding $\rho \approx 3 \times 10^{-6}$ flat to thirty periods on fresh starts.

- A validation chain to the target horizon predicts held-out six-period error at Spearman $+0.98$; the training loss and the one-step error predict it at only $+0.35$, and a one-period chain at $+0.67$.
- **The protocol:** train about ten seeds; chain each on the validation starts to the target horizon; keep the smallest. Selection beats the median seed by a factor 3 to 96.
- The far-horizon check separates the front-runners: chained to thirty periods from twenty fresh starts, the 4×32 choice moved from 3.1×10^{-6} to 3.4×10^{-6} , a factor 1.1, while the 4×64 , 4×128 and 4×50 choices grew by factors 11.3, 8.7 and 24.
- **Caveat carried forward:** a six-period validation window does not guarantee thirty periods for every candidate. The protocol's window must match the horizon the map will be used at.

The reported network is the 4×32 seed-9 map, 3858 parameters.

Chaining beats one large step, and physics buys data-independence rather than accuracy

whole trajectories, relative L_2			
T	chained, 4×50 seeds	chained, selected 4×32	continuous-time, plain
3	2.6×10^{-3}	1.6×10^{-3}	3.8×10^{-5}
7	4.8×10^{-3}	2.1×10^{-3}	6.6×10^{-4}
14	9.0×10^{-3}	2.1×10^{-3}	0.79
27	1.8×10^{-2}	2.1×10^{-3}	0.90
40	2.6×10^{-2}	2.0×10^{-3}	0.94

Pooled over three seeds and 108 test starts; single network for the selected column. The continuous-time column is a *dedicated* fit to one start at $T \leq 7$.

The unselected seeds compound gently, a factor 10 over six periods. The selected network's column is **flat**, a factor 1.3 from half a period to six, and twelve times better than the best unselected seed at six periods. At the matched horizon, chaining beats the one-giant-step mode by a factor 10.

Feasibility of the giant step was checked first with the exact solver: at a full period in one step the stage equations stop being reliably solvable even at $q = 64$. A-stability is a linear property; what binds is nonlinear solvability.

physics against perfect labels, matched pair			
n	physics	data	integrator steps for labels
50	9.2×10^{-3}	9.8×10^{-3}	39,950
125	6.3×10^{-3}	3.0×10^{-3}	99,875
250	4.1×10^{-3}	1.8×10^{-3}	199,750
500	3.4×10^{-3}	1.6×10^{-3}	399,500
1000	3.4×10^{-3}	1.7×10^{-3}	799,000
2000	3.6×10^{-3}	1.6×10^{-3}	1,598,000

Identical network, identical seeded initial weights, identical optimiser and budget. Only the loss and n differ. Both flatten from $n \approx 500$, a **stable factor 2.1 apart**, with the physics variant ahead only at $n = 50$.

The last column is $799n$: the labels for one state are eight true stage states plus the endpoint, produced by running the reference across the step at 10^{-3} and halting exactly at each stage time. **The physics variant's reference cost is exactly zero at every n .**

At $q = 8$ the two losses' minimisers differ by 2.4×10^{-8} , five orders below either variant's error, so the gap is not an information gap but a difference in the path the optimiser takes.

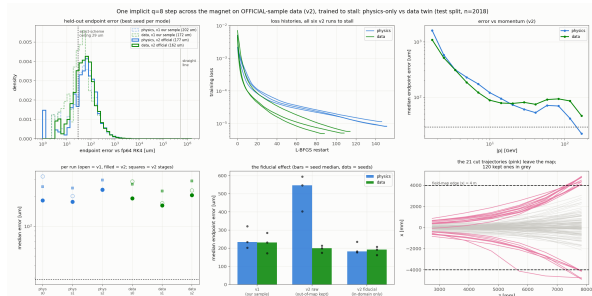
The July baseline: one frozen magnet crossing, and the two losses indistinguishable

frozen leg, $\Delta z = 5177.8 \text{ mm}$, $q = 8$, 4×50

exact scheme (the ceiling)	23 μm
network, label-free physics loss	183 [177, 183, 235]
network, supervised data twin	192 [162, 192, 208]
straight line (ignore the magnet)	520,444

Three things fixed by the end of July, unrevisited since:

- **the leg:** the modal plane pair of the 11,046 forward cross-magnet legs, $z_0 = 2648.2 \text{ mm}$ to $z_1 = 7826.0 \text{ mm}$, the last plane of the upstream tracker to the first of the scintillating-fibre tracker. The hardest of the routine steps and the one where a surrogate is worth most.
- $q = 8$, from the exact-scheme measurement.
- **the first converged label-free result**, with the fiducial requirement in place and trained to a confirmed stall.



Error distributions for the three label-free and three supervised seeds; the training histories per optimiser restart with the confirmation pass marked; and the comparison against the in-house sample the official one replaced.

The point of the figure is the overlap. With the fiducial requirement in place, the label-free loss and the supervised twin are not distinguishable at three seeds on this leg. The factor-two gap the July experiments chased was a budget artefact.

Stage-count sweep: at two and four stages the network reproduces the scheme's own error

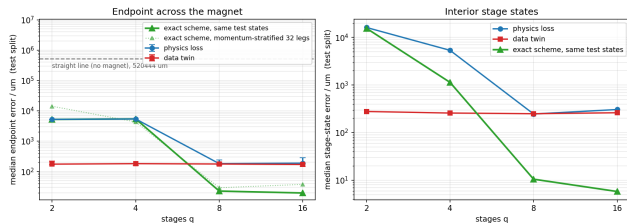
stages q	physics loss	data twin	exact scheme	exact scheme, 95th pct.
2	5202 [5199–5218]	174 [168–223]	5198	37,728
4	5371 [5352–5394]	181 [157–181]	5401	54,057
8	181 [165–240]	176 [162–204]	23	1610
16	187 [186–280]	170 [147–176]	20	325

Frozen cross-magnet leg, 4×50 , three seeds each, 24 runs, all converged. Test-split median endpoint error in μm , median over seeds with the seed range in brackets. The ceiling is the same q -stage scheme solved by root-finder on the very same 2018 test states. Straight line: 520,444 μm at every q .

- At $q = 2$: 5202 against 5198, **within 0.1 percent**. At $q = 4$: 5371 against 5401, **within 0.6 percent**.
- The network is not failing. It is solving the equations it was given, and those equations are 5 mm wrong on this leg. This is the cleanest available demonstration that the physics loss drives the network onto the solution of the discrete scheme, whatever that solution happens to be.
- The twin at the same q sits at 174 and 181 μm , thirty times better, because its labels are the true trajectory and it never sees the scheme at all.
- Between $q = 4$ and $q = 8$ the ceiling crosses *below* the network's own floor. From there the network and the optimiser limit, and $q = 16$ agrees with $q = 8$ within the seed spread while costing 68 outputs per sample instead of 36. That fixes $q = 8$.
- **Two surprises.** $q = 4$ is no better than $q = 2$ and slightly worse at the ceiling: formal order $2q$ is asymptotic in small steps, and one 5178 mm step through a magnet is nowhere near that regime. And $q = 16$ is not the hard case to train, 113 to 115 restarts against 154; the hardest optimisation in the grid is $q = 4$.
- The root-finder converged below 10^{-8} on 99.9 percent of legs at every q , so these are genuine solutions of the discrete equations.

The same sweep as a picture, and why a ceiling must be quoted on the population being scored

Stage-count sweep on the frozen magnet crossing (median over 3 seeds, bars = seed spread)



Left: endpoint error against q . Right: the same measure over the eight interior stage states. Blue is the label-free loss, red the twin, solid green the exact scheme on the very same test states, dotted green the same scheme on the momentum-stratified sample of 32 legs. Bars span three seeds; the dashed horizontal line is the straight-line baseline.

Blue and green lie on top of one another at $q = 2$ and $q = 4$, which is the physics loss reproducing the scheme's own error, and they separate at $q = 8$ where the scheme's error drops below the network's floor. The red curve is flat at about $175 \mu\text{m}$ at every q , because the twin never sees the scheme.

The $29 \mu\text{m}$ correction. An earlier measurement of the exact scheme, made on 32 legs drawn stratified in momentum and each on its own start plane, gave $29 \mu\text{m}$ at $q = 8$, and that number was quoted in July. Stratifying in momentum deliberately over-weights the soft, hard-bending tracks, so it describes a harder population than any test split used here.

On the 2018 frozen-leg test states with their natural momentum mix the ceiling is $23 \mu\text{m}$, and the pre-stated ceilings of about 14 mm at $q = 2$ and 4.3 mm at $q = 4$ become 5198 and $5401 \mu\text{m}$. Both curves are carried in the figure so the difference is visible rather than argued about. The qualitative claim survives intact.

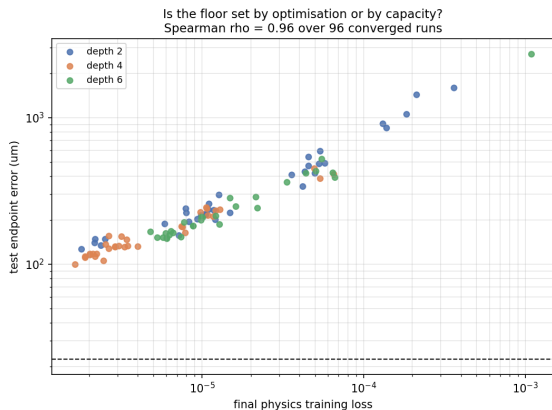
The full architecture grid: twelve architectures, ten seeds, 130 runs

loss	depth $\times$ width	parameters	converged	median	best seed	worst seed	median final loss
physics	2 $\times$ 32	2,436	5/10	1059	853	1600	1.83×10^{-4}
physics	2 $\times$ 50	4,686	9/10	470	340	593	4.54×10^{-5}
physics	4 $\times$ 32	4,548	3/10	411	387	450	5.37×10^{-5}
physics	6 $\times$ 32	6,660	4/10	427	391	2725	6.51×10^{-5}
physics	4 $\times$ 50	9,786	10/10	222	165	244	1.06×10^{-5}
physics	2 $\times$ 100	14,336	10/10	223	201	299	1.08×10^{-5}
physics	6 $\times$ 50	14,886	6/10	326	248	525	2.74×10^{-5}
physics	4 $\times$ 100	34,536	10/10	133	128	156	3.04×10^{-6}
physics	2 $\times$ 200	48,636	10/10	153	128	241	4.19×10^{-6}
physics	6 $\times$ 100	54,736	10/10	191	152	242	9.33×10^{-6}
physics	4 $\times$ 200	129,036	10/10	115	100	134	2.06×10^{-6}
physics	6 $\times$ 200	209,436	9/10	159	150	208	6.05×10^{-6}
data twin	4 $\times$ 200	129,036	10/10	128	105	159	1.34×10^{-6}

Frozen cross-magnet leg at $q = 8$, ordered by parameter count. Test-split median endpoint error in μm over the converged seeds. **Unconverged runs are never pooled.** The 24 label-free runs that did not confirm are all at width 32 or depth 6, and none of them diverged: they failed the confirmation pass, in which a fresh optimiser moves a small or deep network again rather than re-stalling.

Depth 4 is the optimum and it is interior. Depth 6 is never better and is 39 to 47 percent worse at widths 50, 100 and 200; it is also much harder to train, 4, 6, 10, 9 seeds converging against 3, 10, 10, 10 at depth 4. Depth 2 is worse at every width, by 33 percent at width 200 rising to 158 percent at width 32.

The floor is set by the optimiser and not by capacity, and the evidence is one band

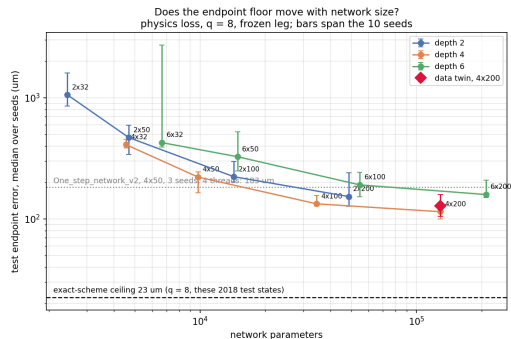


Final training loss against test-split endpoint error, both axes logarithmic, one point per converged label-free run, coloured by architecture.

- Over all **96 converged label-free runs** the Spearman rank correlation between final training loss and test endpoint error is **+0.965**. Three decades of loss map monotonically onto one and a half decades of error.
- **If the floor were a capacity limit**, the small architectures would sit above the band at their own loss values, forming separate branches. They do not. The endpoint error is a readout of how far the optimiser got, not of what the network could represent.
- A wider network helps only because L-BFGS gets further down on it: the 4×50 networks stall at a median final loss of 1.06×10^{-5} , the 4×200 networks at 2.06×10^{-6} .
- **The prediction that follows**, and which the residual redesign then tested: anything improving the *conditioning* of the optimisation problem should pay more than more parameters do.

This is the same number, to the second decimal place, that the van der Pol study measured on a two-dimensional autonomous oscillator with an analytic right-hand side. I did not expect two systems with that little in common to agree that closely, and I have no explanation for it beyond the shared optimiser.

Returns collapse with size, and at the largest size the label-free loss overtakes its twin



Median test endpoint error against parameter count, both axes logarithmic. Each point is one architecture; the three curves are depths 2, 4 and 6; bars span ten seeds. The red diamond is the supervised twin at 4×200 . The dotted line is the July three-seed baseline at $183 \mu\text{m}$; the dashed line at the bottom is the exact-scheme ceiling at $23 \mu\text{m}$ on the same 2018 test states.

- Growing from 2436 to 129,036 parameters, a factor 53, moves the median from 1059 to $115 \mu\text{m}$, a factor 9.2. The last width doubling buys **14 percent for 3.7 times the parameters** and 1.8 times the wall time.
- The depth-4 curve flattens above about thirty thousand parameters, and **the gap between the flattening curve and the ceiling is the factor 5** this study attributes to the optimiser.
- **At 4×200 the label-free loss beats its own supervised twin:** $115 \mu\text{m}$ [100–134] against $128 \mu\text{m}$ [105–159], so the twin's median is 12 percent higher. The twin also has the wider seed spread, a factor 1.51 between best and worst against 1.33, while reaching its stall in about half the restarts. The factor-two gap the July experiments chased is gone.
- **Seeds must be pooled.** Over architectures with at least three converged seeds the median luckiest-to-unluckiest ratio is 1.54, shrinking with size to 1.22 at 4×100 .
- **Validation selection is safe for the one-step task:** rank correlation between validation and test endpoint error is $+0.995$ over the 96 runs, the validation-best seed is also the test-best in 9 of 12 architectures, and where it is not the penalty is at most 2.3 percent. **That stops being true the moment the network is chained.**

The recommendation this produced: 4×100 as the smallest adequate network, with 4×200 as the check that a result is not size-limited. 4×50 sits on the steep part of the curve, where a result reports the network size as much as the physics.

Training where no labels exist: the reversed magnet polarity reaches the same floor

quantity	magnet up (reversed)	magnet down (original)
network, label-free physics loss	229 [171–296], 10 seeds	222 [165–244], 10 seeds
network, supervised data twin	204 [187–230], 3 seeds	192 [162–208], 3 seeds
exact-scheme ceiling on the test states	22.1	22.5
straight line	525,184	520,444
network divided by its own ceiling	$10.4 \times [7.7\text{--}13.4]$	$9.8 \times [7.2\text{--}10.8]$
fiducial removals, train / validation / test	0 / 0 / 0	21 / 20 / 15
exact scheme, worst case	38 mm	258 mm

Design. Identical 4×50 network, identical protocol, identical training population; the single change is that the loss is built on the reversed map. Ten label-free seeds plus three supervised controls, 13 runs, all converged. The reversed-polarity references exist only to score; the label-free runs never see them.

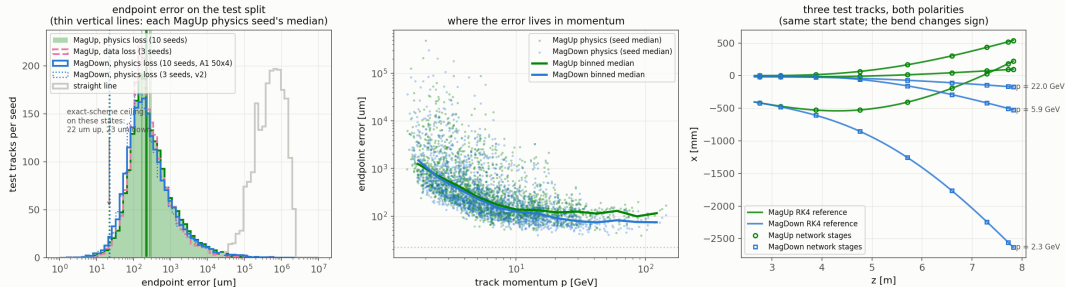
Three checks before submitting. The two maps are exactly one another's negative: on 200,000 random in-map points the maximum of $|\mathbf{B}_{\text{up}} + \mathbf{B}_{\text{down}}|$ is 0.0 in all three components. Both polarities are built from the same event rows, since the selection uses no field. And a probe run established that the polarity flag really reaches the loss: one restart from the same seed gives 2.62×10^{-3} reversed against 8.03×10^{-3} original.

The two ten-seed ranges overlap almost completely, and the four percent difference in medians is far inside a seed spread that is itself a factor 1.7 wide reversed and 1.5 original. Both are three orders of magnitude below the straight line, so **the network learned the bending and not merely the geometry, and it learned it with the correct sign.**

One genuine asymmetry, and it is in the tracks rather than the map. The fiducial requirement removes 21, 20 and 15 states on the original polarity and none on the reversed one: the sample's charge and angular distribution is not symmetric in x , and the original polarity is the one that bends these particular soft tracks out of the map while the reversed bends them back in. A control re-scoring the reversed networks on only the tracks the original also keeps moves the median from 229.4 to 226.6 μm , about one percent, so medians are unaffected. Statements about *tails*, however, must say which polarity.

The reversed polarity in pictures: same distribution, same momentum curve, opposite bend

A4 - the one-step network trained with no labels on the magnet-up field



Left:

the distribution of test-split endpoint errors on a logarithmic axis, with the ten reversed-polarity label-free seeds in filled green, the three reversed supervised seeds dashed, the ten original-polarity label-free seeds solid blue, the three July original seeds dotted, and the straight-line baseline in grey near one million micrometres. The two polarities lie on top of one another and the two ceilings, 22 and 23 μm, are marked at the left. Middle: endpoint error against track momentum with binned medians overlaid; the two polarities trace the same curve, falling from about one millimetre at 2 GeV to about 100 μm above 20 GeV. Right: three test tracks propagated from the same start states under both polarities, with the network's eight stage states as markers on the reference curves. The trajectories curve in opposite directions, which is the check that the network learned the sign of the bending.

One network for every kind of step, absolute outputs: worse than a straight line on short steps

leg	physics loss			data twin			straight	ceiling
	4×50	4×100	4×200	4×50	4×100	4×200		
A vertex fetch, 341 mm	849	404	229	653	385	227	10.9	0.0013
B cross-magnet, 5175 mm	6742	2600	1740	2395	933	777	444,075	46
C plane-to-plane, 70 mm	535	300	213	486	235	182	6.6	6×10^{-7}

Median μm over converged seeds. The leg becomes an input: each sample carries its own z_0 and Δz , normalised, and the stage planes become per-sample quantities $z_0 + c_j \Delta z$.

Training size fixed by a rule stated before any result was seen: the largest of 2000, 4000, 8000, 16000 for which one restart runs under 90 seconds, giving $N = 8000$, or 7935/3969/3961 states after the fiducial cut. Leg mix is the training set's own and is not stratified: 1317 A, 1060 B, 5558 C. Ten label-free seeds and three supervised at 4×50 and 4×100 (26 runs, all converged); a second wave at 4×200 (13 runs, 12 converged).

- One label-free network does serve all three leg types, and nowhere near any of their ceilings. Against the frozen-leg network on the same kind of leg, making the leg an input **costs a factor 14**, and a factor 11.7 against the ten-seed frozen figure of $222 \mu\text{m}$ even though this network is twice as wide.
- On the two short types **the network is worse than doing nothing**, and the widest is still 32 times (C) and 21 times (A) worse than ignoring the magnet.
- Width keeps paying with clear diminishing returns: leg B goes $6742 \rightarrow 2600 \rightarrow 1740$, factors 2.6 then 1.5, still 38 times above its $46 \mu\text{m}$ ceiling.
- **The twin is uniformly ahead here**, by a factor 2.8 on leg B at 4×100 , where on the frozen leg the two were indistinguishable. The gap opens exactly where the leg geometry varies.

The hypothesis that experiment produced: it is the output parameterisation, not the capacity

Stated before testing. The general-leg network is limited by predicting the *absolute* state at each stage, scaled by a single population-wide constant per component. Four pieces of evidence:

1. The output scale on the general dataset is 602 mm in x and 341 mm in y , set by the whole population and dominated by the 5 m crossings. The correction the network must produce on a 70 mm hop is about $7\text{ }\mu\text{m}$.
Resolving $7\text{ }\mu\text{m}$ on an output whose scale is 602 mm is one part in 10^8 . On the frozen dataset the scale is 396 mm and the correction 520 mm: a completely different regime.
2. The error is roughly a fixed fraction of the largest step in training, not of each step: 404, 2600, 300 μm on legs whose true corrections span five orders of magnitude.

3. On the short legs the error is *flat* across the eight Gauss nodes, the signature of an offset. On the long leg it grows monotonically to the endpoint, roughly doubling, which is a bend being under-resolved.
4. The frozen-leg network, same architecture, loss, optimiser and data source but one geometry and therefore a matched scale, is a factor 14 better on that geometry.

The alternative was capacity, and the second wave separated them. The size study predicted width would buy about 14 percent over 4×100 on the frozen leg; on the general dataset it bought a factor 1.5 on leg B, **leaving the factor 14 largely intact and the short legs still an order of magnitude worse than a straight line.** That is consistent with the parameterisation hypothesis and not with the capacity one.

Chaining the absolute-output network along real particle paths: the error doubles per leg

legs walked	physics loss			data twin			particles
	4×50	4×100	4×200	4×50	4×100	4×200	
1	440	246	174	388	217	165	4563
2	1423	783	540	1275	652	510	4563
3	2562	1443	1029	2225	1051	894	4563
4	3919	2293	1738	3634	1484	1490	4563
5	8163	4831	4133	6991	2869	2972	2914
6	14,479	8904	8659	10,087	4820	5695	1481
7	15,999	10,248	10,450	10,739	5283	6476	853

Median endpoint error against the reference path, test split, in μm . All 4563 particles contribute up to leg 4, so the first four rows are the like-for-like comparison. Construction: collect the planes named by each test particle's stored forward legs, sort them, call consecutive planes a leg. 4563 particles with four or more legs, 23,500 legs, median leg length 70 mm, 99th percentile 5.2 m. The obvious rule of butting rows together is unusable: only 15 of 7636 test particles have four rows that butt exactly.

Nothing breaks and nothing saturates. No chain diverges and the 95th percentile grows at the same rate as the median. Step-to-step growth ratios at 4×100 are 3.2, 1.8, 1.6, 2.1, 1.8, 1.1, essentially the same at every architecture and both losses. **That is not the square-root growth of independent random errors; it is systematic,** each leg starting from an already displaced state and adding its own bias. Over a full path the network is 5 to 10 mm out, and width shifts the curve down without changing its slope.

Chaining also changes how a seed must be chosen. The cheap one-step validation error, an excellent selector for the one-step task, carries *no* information about chained performance: rank correlation 0.08 at 4×100 , and the best one-step seed is only sixth best when chained. The validation *chain* error transfers almost perfectly, 0.92 and 0.95 at the two architectures and 0.94 at 4×200 . A seed must be selected on the quantity that will actually be used, which is the same lesson the oscillator taught.

Leg D: replacing a hard step by two easier ones makes it worse, under both designs

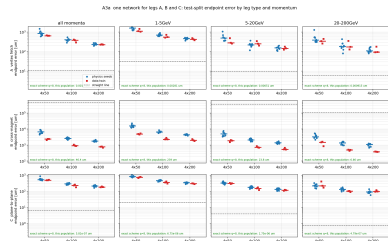
	absolute outputs			straight-line residual		
	physics 4×100	twin 4×100	physics 4×200	physics 4×100	twin 4×100	physics 4×50
two composite steps	59,416	33,416	35,034	4252	2374	4768
the same leg in one giant step	8778	3158	4563	3749	2213	3806
exact $q = 8$ scheme on these legs: 752; $q = 16$: 216; straight line: 948,000						

Median μm over 2726 test particles. Leg D is the single backward step from the first post-magnet plane to the collision vertex, median length -8.6 m. **No leg of this geometry was in any training set**, so this is an out-of-training test by construction. The stored label agrees with a fresh reference pass to $0.045\ \mu\text{m}$.

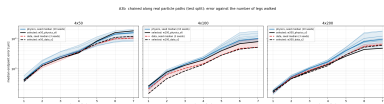
Under absolute outputs. Composite stepping is four to ten times worse than the single giant step, and both are more than an order of magnitude above the ceiling. The reason is visible in the first hop, which is longer than any leg in training and is already 1.5 to $10\ \text{mm}$ out. **This is not a sign problem**: the network has seen backward steps, since all 1317 A legs and 490 of the 1060 B legs are backward. It is that no leg-D geometry was in training at all. It does still beat the straight line by a factor 100 to 300.

Under the residual design. The composite improves by a factor 14, from 59.4 to $4.3\ \text{mm}$, far more than the single step's factor 2.3: the composite's first hop is a -6.4 m backward step, longer than any B leg, and it went from 4.1 to $1.5\ \text{mm}$, while its second hop is vertex-fetch-like, exactly where the residual arm gained a factor 92. **The gap has closed from a factor 6.8 to 1.13**, so the earlier conclusion that composite stepping actively destroys accuracy is now only marginally true. Both remain five times the ceiling, and the 95th percentile barely moved, 142 against $149\ \text{mm}$.

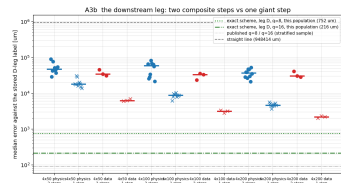
The absolute-output network in pictures: by leg, chained, and on the giant step



Endpoint error by leg type (rows: A vertex fetch, B cross-magnet, C plane-to-plane) and momentum band (columns: all momenta, then 1–5, 5–20 and 20–200 GeV). Blue: ten label-free seeds. Red: three supervised seeds. Dashed grey: the straight line. Green: the exact-scheme ceiling on that same population. **On the top and bottom rows the straight line lies below every network point.**



Median endpoint error against the number of legs walked. Each curve is one architecture and loss; bands are the seed spread, dashed curves the 95th percentiles. The contributing-particle count falls after leg four, marked on the axis. **The curves are straight on this axis over the first four legs, which is geometric growth:** the error roughly doubles per leg rather than growing like the square root of the number of legs.



Leg D, the single backward step of median length —8.6 m from the first post-magnet plane to the collision vertex, a geometry that appears in no training set. Left: the error distribution for the composite route and for the single giant step, with the ceiling and the straight line marked. Right: the same against momentum. **The composite route, which was supposed to replace a hard step by two easier ones, is worse than the single step at every momentum, because its first hop is longer than any leg in training.**

The straight-line-residual redesign: the two short legs transform, the magnet crossing does not

leg	absolute, physics		residual, physics		residual		
	4×100	4×200	4×100	4×50	twin	straight	ceiling
A vertex fetch	404 [286–538]	229 [201–277]	4.4 [2–6]	4.1 [3–6]	0.058	10.9	0.0013
B cross-magnet	2600 [2106–3152]	1740 [1632–2154]	1026 [878–1209]	1136 [945–1627]	512	444,075	46.4
C plane-to-plane	300 [252–335]	213 [165–235]	1.6 [1–2]	1.2 [1–1]	0.012	6.6	5.8×10^{-7}

whole test split, all leg types pooled; the straight line on this split is $16.1 \mu\text{m}$

output design	physics loss			data twin		
	4 × 50	4 × 100	4 × 200	4 × 50	4 × 100	4 × 200
absolute states	721	397	273	637	321	237
straight-line residual	2.12	2.64	not run	0.06	0.04	not run

Exactly one thing changed: what the network outputs. The dataset is byte for byte the same population plus three scale arrays per split; the physics loss, the optimiser protocol, the stage count, the splits, the momentum bands, the scorer and the ceilings are all unchanged. The one deliberate consequential change is that the twin's normalisation follows, dividing by the per-sample residual scale, because otherwise it would be trained on an objective the physics arm is not. 26 runs, all 26 converged.

Reading the redesign across, and what the untrained network already scores

1. **The two short legs are transformed.** Leg A $404 \rightarrow 4.4 \mu\text{m}$, a factor 92; leg C $300 \rightarrow 1.6 \mu\text{m}$, a factor 190. Both now sit *below* the straight line, by 2.5 and 4, where the absolute design sat thirty to fifty times above it. That is the hypothesis's own prediction.
2. **The cross-magnet leg improves by only a factor 2.5,** $2600 \rightarrow 1026 \mu\text{m}$. Real, but an order of magnitude short of the same transformation. Leg B is the one leg where the absolute scale was roughly right, so it had least to gain. The redesign fixed a conditioning problem, and what is left on leg B is not one.
3. **Width has stopped mattering** on A and C: 4×50 and 4×100 are within 20 percent, and the *narrower* network is the better of the two on both. The absolute-output verdict read the width trend as evidence that width was the binding constraint. It was never width.
4. **The twin is extraordinary on the short legs,** 0.058 and $0.012 \mu\text{m}$, four orders better than the absolute twin on each, and still twice as good as the physics loss on leg B.
5. **Distance to the ceiling, the honest scorecard.** The residual 4×100 physics network is 3400 times above the ceiling on A, 22 on B and 2.7×10^6 on C. The absolute factors were 310,000, 56 and 5×10^8 . **The network is nowhere near the scheme it is solving on any leg,** and on the short legs the scheme is so nearly exact that no network will reach it.

The initialisation check, run before any farm time. The straight-line term agrees with an independent construction to $4.5 \times 10^{-13} \text{ mm}$; zeroing the last layer makes output minus straight line exactly 0.0 mm ; at the trainer's own initialisation the raw output has median 0.057 in units of the residual scale, which is $0.51 \mu\text{m}$ at the median endpoint. **An untrained residual network already scores $15.5 \mu\text{m}$ on the whole test split, against 16.1 for a straight line,** where the converged absolute networks scored about 300. *And the optimisation got easier:* stalls at 40 to 64 restarts against 84 to 121, at a loss an order of magnitude lower; three of 26 needed resubmission against 20 of the first 26.

Where the error lives inside the step, and the tails that did not follow the medians

	node 1	2	3	4	5	6	7	8	endpoint
absolute, leg A	365	340	353	320	312	408	349	381	404
residual, leg A	0.06	0.10	0.43	1.01	1.74	2.81	3.64	4.34	4.42
absolute, leg C	289	240	249	270	260	271	268	278	300
residual, leg C	0.03	0.05	0.29	0.65	0.88	1.12	1.34	1.52	1.57
absolute, leg B	987	905	1054	1436	1907	2326	2536	2519	2600
residual, leg B	108	143	277	456	616	812	924	988	1026

Median position error at each Gauss node and at the endpoint, test split, 4×100 physics loss, median over seeds, μm .

This is the clearest single piece of evidence that the diagnosis was right. A flat profile means the error is present before any physics has happened, so it is an offset. A rising profile means it is accumulated along the step. The absolute error on A and C is flat; the residual error rises by a factor 50 to 70 from a very small first node. Leg B kept its monotonic growth and simply scaled down by 2.5.

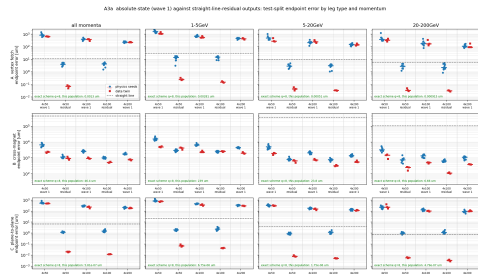
95th percentiles, 4×100 , all momenta				
leg	abs. physics	resid. physics	abs. twin	resid. twin
A	30,693	23,640	10,445	20,128
B	41,542	27,080	18,395	17,425
C	2862	292	2193	32

A small population got worse while the bulk got thousands of times better. Those are the soft tracks that bend hardest.

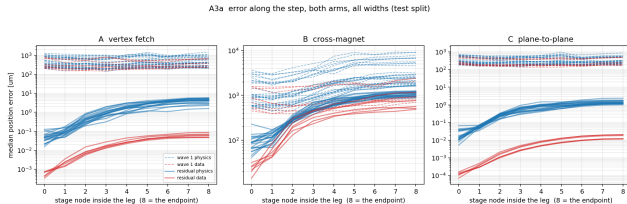
The residual scale is a first-order estimate, and where the true deviation is many times that estimate the network is back to writing down a large number. The residual twin's tail on leg A is twice the absolute twin's.

Fixing the median did not fix the tail, and **the tail is what an extrapolator with a bounded time budget actually has to survive.**

The redesign in pictures: crossing the straight line, and an offset becoming an accumulation



The same layout as the absolute-output figure, with the residual arm added. Rows are leg types, columns momentum bands. The absolute-output seeds are the upper cloud in each panel and the residual seeds the lower one. **On the top and bottom rows the residual cloud crosses from above the straight line to below it**, which is the whole point of the redesign; on the middle row it moves down by a factor 2.5 and stays well above both the ceiling and its own twin. The improvement on the short legs is uniform in momentum, a factor 46 to 200 in every band, so it is not an artefact of the momentum mix.



The error along the step, both output designs, one panel per leg type. Horizontal axis: the eight Gauss–Legendre nodes and then the endpoint. Vertical axis: median position error in micrometres, logarithmic.

A flat profile means the error is present before any physics has happened, so it is an offset the network cannot resolve at the population scale. **A rising profile means it is accumulated along the step**. The absolute-output curves on the two short legs are flat; the residual curves rise steeply from a very small first node. The cross-magnet leg kept its monotonic growth in both designs and simply scaled down by 2.5, which is the signature of a bend still being under-resolved rather than of an offset.

The one cell where the residual network is still worse than a straight line is plane-to-plane at 20 to 200 GeV, 1.2 against $0.74 \mu\text{m}$: a stiff 70 mm hop is so nearly straight that there is essentially nothing to predict, and the network's own noise floor is above the thing it is correcting.

Chaining the residual networks: better everywhere, and compounding faster

legs	absolute 4×100		residual			
walked	physics	twin	4×100	4×100 twin	4×50	particles
1	246	217	1.2	0.01	1.1	4563
2	783	652	60	2.7	59	4563
3	1443	1051	281	32	261	4563
4	2293	1484	572	118	568	4563
5	4831	2869	1750	471	1919	2914
6	8904	4820	3804	1052	3973	1481
7	10,248	5284	4492	1259	4739	853

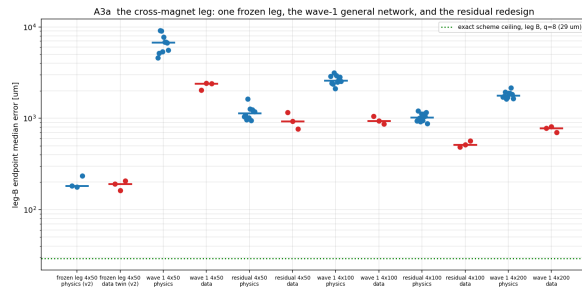
The chain is better everywhere. After four legs the residual network is 4.0 times better with the physics loss, 572 against 2293 μm , and 12.6 times better with the twin. Over a full seven-leg path it is still 2.3 and 4.2 times better. The 95th percentile improves by the same factor at every step, 13.1 against 38.6 mm at leg four, so this is the whole distribution moving.

But the compounding got worse. Growth ratios at 4×100 with the physics loss: 48.6, 4.67, 2.04, 3.06, 2.17, 1.18, against 3.18, 1.84, 1.59, 2.11, 1.84, 1.15 for the absolute design. The residual network starts two hundred times lower, at essentially its own plane-to-plane number, then multiplies by 49 on the second leg.

The reason is visible in the one-step table. A chain's early legs are short hops, where the residual network is at 1.6 μm ; its later legs include the cross-magnet crossing, where it is at 1026 μm , the leg the redesign barely improved. **The chain error is dominated, after two or three steps, by the one leg that did not get better.** The redesign bought a much better start, not a slower compounding.

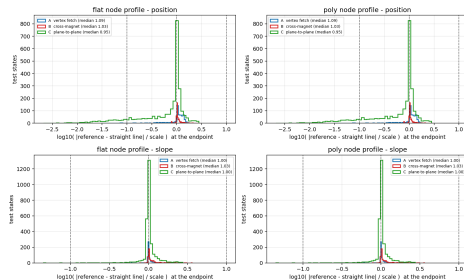
Seed selection survives: ranking physics seeds by validation chain error at four legs and reading out on test gives 0.98 and 0.90. The cheap alternative improves but is still not usable, 0.72 at 4×100 and 0.35 at 4×50 , where in the absolute arm it carried no information at 0.08. Ten seeds differ by 60 percent in chained error at fixed architecture.

The cross-magnet leg across all three designs, on one axis



The frozen-leg network, which trains on that geometry alone; the general network with absolute outputs; and the general network with the straight-line-residual parameterisation. The exact-scheme ceiling on each population and the straight-line baseline are drawn.

Residual redesign: is the deviation from a straight line $O(1)$ on its own scale? (dashed: the 0.1-1.0 band; chosen: flat)



The

check that had to pass before any farm time was spent: the distribution, by leg type, of the ratio the network actually has to emit, that is the reference deviation from a straight line divided by the label-free scale, for both node profiles. Medians 1.09, 1.03, 0.95, so the pre-fixed rule selected the flat profile. The same label-free expression lands within ten percent of the truth on a 341 mm backward vertex fetch, a 5.2 m magnet crossing and a 70 mm sensor hop.

Making the leg an input costs a factor 14; the residual redesign recovers a factor 2.5 of it. This gap, and not the output scale, is what the next experiments attack.

The whole programme on one page

	frozen leg		magnet up	one network, legs A / B / C, 4×100		chained
	4×50	4×200	4×50	absolute	residual	4 legs
exact scheme (ceiling)	23	23	22	0.0013 / 46 / 6×10^{-7}	0.0013 / 46 / 6×10^{-7}	n.m.
network, physics loss	222	115	229	404 / 2600 / 300	4.4 / 1026 / 1.6	572
network, data twin	192	128	204	385 / 933 / 235	0.058 / 512 / 0.012	118
straight line	520,444	520,444	525,184	10.9 / 444,075 / 6.6	10.9 / 444,075 / 6.6	n.a.
seed range, physics	[165–244]	[100–134]	[171–296]	10 seeds	10 seeds	selected seed

Test-split median endpoint error in μm . Each column's ceiling is measured on that column's own population. The evidence base is 232 training runs; 208 reached a confirmed stall, the 24 that did not are all in the architecture grid at width 32 or depth 6, and **no run blew up**.

Question 1, does it work: yes, on every test. Training is stable; the network sits on the scheme's error to 0.1 and 0.6 percent where the scheme limits; the shortfall where the network limits is identified as optimisation at +0.965 over 96 runs; at 4×200 it beats its own twin; and it reaches the same floor on a polarity with no labels.

Question 2, can it be used: the redesign changed three of four answers. Short legs went from thirty to fifty times worse than ignoring the magnet to two to four times better; the split went $274 \rightarrow 2.1 \mu\text{m}$ against 16.1; the chain improved by 4 to 13 at every length; composite stepping went from six times worse to 13 percent worse. **What is left is one leg.**

The cost statement, flagged. A median of 44 root-finder residual evaluations per leg against one forward pass. That is an operation count, not a measurement.

Two ceilings and a floor, and the gap between them

1. The C^0 field map caps the scheme at $23\mu\text{m}$

- The map is a measured table on an $81 \times 81 \times 146$ grid at 100mm spacing, read by trilinear interpolation. The interpolant is continuous; its derivative is not. At every cell boundary the gradient jumps.
- High-order schemes derive their order from Taylor expansions of the right-hand side, which require it to be differentiable. It is not. A 5178mm crossing passes about 52 cell boundaries in z alone, and many more in x as the track bends.
- So the ceiling falls from $5401\mu\text{m}$ at $q = 4$ to $23\mu\text{m}$ at $q = 8$ and then *stops*, reaching only $20\mu\text{m}$ at $q = 16$: the median moved 13 percent while the unknowns doubled. The tail improved much more, $1610 \rightarrow 325\mu\text{m}$, consistent with the tail being dominated by soft tracks that sample the map most aggressively.
- **This is not a property of any network.** It is what the mathematics being solved is worth on this field map.

2. The network floor sits a factor 5 above that, and it is the optimiser: three decades of training loss map monotonically onto one and a half decades of error, and every architecture and every seed lies on one band.

3. The binding constraint is one leg.

- The magnet crossing inside a shared network sits at $1026\mu\text{m}$ against a $46\mu\text{m}$ ceiling on its own population, a factor 22, and the redesign moved it by only 2.5.
- It is not width: 4×50 and 4×100 agree within ten percent on it. It is not the output scale: leg B is the one leg where the absolute scale was roughly right.
- **The supervised twin is also stuck**, at $512\mu\text{m}$, a factor 11 above the ceiling, while reaching 0.058 and $0.012\mu\text{m}$ on the other two legs. Whatever limits leg B limits both losses, so it is not a property of the label-free objective.
- Cross-magnet legs are **13 percent** of the training states; plane-to-plane hops are 70 percent. Under the absolute parameterisation the population-wide scale was set by the long legs and they dominated the loss anyway. Under the residual parameterisation every leg is order one, so the gradient is dominated by the easy majority. **The redesign fixed the conditioning and, in doing so, handed the optimisation budget to the plane-to-plane hops.**

What I cannot claim yet

- **Chaining compounds.** Roughly a factor two per leg in every architecture and both losses, and the residual design compounds *faster* than the design it replaced. Nothing diverges, nothing saturates, and width shifts the curve down without changing its slope.
- **The tails did not improve with the medians.** On legs A and B the residual 95th percentile is no better than the absolute one, and the residual twin's tail on leg A is twice as large. A 95th percentile of 27 mm on the magnet crossing is not usable no matter what the median is.
- **No throughput benchmark exists.** Everything runs in double precision on one core. Nothing here says what the same network does in reduced precision on a graphics processing unit, which is where it would have to live.
- **Labels are free at LHCb.** A reference label is one integrator call away, so the label-free advantage is *methodological* rather than economic: portability across field maps, geometries and conventions. The reversed-polarity test is the demonstration of that, and it is the honest form of the claim.

Harness lessons, recorded so they cost someone else less than they cost me

- **The restart cap counted the stall phase and the confirmation pass together,** so runs that stalled near the cap were recorded as unconverged when they were about to converge. Three experiments hit this independently on the same day. The default is raised from 150 to 400; the cleaner fix, giving the confirmation pass its own budget, is not yet applied. No result changes, because a larger cap cannot affect a run that stopped on the stall criterion below it.
- **Ceilings must be quoted on the scored population.** Mixing the stratified and natural conventions turns a factor 56 into 90 or 36, and neither means anything.
- **Single thread, and the boundary it creates.** Multithreaded PyTorch spin-waits on these nodes and is about 68 times slower. September runs are single-threaded and July runs used four threads; a different reduction order changes the last bits and 200 line-searched iterations amplify that into the fourth significant digit. Comparisons across that boundary are made at the level of seed ranges, never digit for digit.
- **One unconverged seed** in the second wave, at 111 restarts of its 400 cap, resubmitted but caught by a long queue. Reported unconverged and excluded from every median; it ranks third on the validation chain and does not affect the selection.

What would falsify the edge: two tests with their criteria stated in advance

Test 1: the throughput benchmark

Design. The 4×200 residual network in single precision, batched, against the production extrapolator, on the deployment hardware, over the same test population, with both timed end to end including the field lookups. Accuracy matched by tuning the production integrator's step until its median endpoint error equals the network's on that population, so the comparison is not made at a different accuracy on each side.

Pass. The network is faster at matched accuracy, and its per-batch time has a variance small enough to schedule against, with the 99th percentile within a stated factor of the median.

Fail. The network is not faster at matched accuracy; or reduced precision costs more accuracy than the accuracy budget allows; or the fixed-cost claim does not survive batching, because the production integrator amortises better than the operation count suggests.

Either way this is the decisive result, and it is currently the largest unmeasured quantity in the programme.

Test 2: the cross-magnet-only network

Design. Train the residual parameterisation on leg B alone, with the start plane and step length still inputs, so the network must serve a *family* of magnet crossings rather than one frozen geometry. Same protocol, same seeds, same ceiling measured on the same population. Its cheaper twin, a stratified training mix over all leg types, answers the same question from the other side.

Pass, meaning the minority-share explanation is right. It recovers most of the factor 22, landing within roughly a factor 3 of the $46 \mu\text{m}$ ceiling. Remedy: a training mix, which is a one-line change to the dataset builder.

Fail, meaning the leg is genuinely harder. It stays near $1026 \mu\text{m}$. Then a 5.2 m crossing needs more than eight stages or more than a first-order scale, and the remedy is in the parameterisation or the scheme, not in the data mix.

The two outcomes point at different work, which is why this is worth running before anything else on the accuracy side.

And the standing falsifier for the whole product claim: a 95th percentile the trigger cannot budget for. At 27 mm on the magnet crossing, a median improvement changes nothing about deployability.

The rest of the queue, in order

1. **A smoother field surrogate.** The $23\text{ }\mu\text{m}$ ceiling exists because the interpolated map has no continuous derivative, so past $q = 8$ there is no smoothness left for extra order to exploit. A smooth surrogate for the field, fitted once offline, lifts the ceiling for the scheme and for every network trained against it, and removes the reason high order currently buys nothing. It also changes what is being solved, which is why it is a question for this room and not a decision for me.
2. **A second-order residual scale.** The first-order estimate is right to ten percent at the median and wrong by a large factor on exactly the soft, hard-bending tracks that dominate every tail in this work. Folding in the curvature of the straight-line path through the field, rather than only the field magnitude along it, stays label free because everything it needs is still an input.
3. **Two small items the campaign produced.** The confirmation pass should have its own restart budget rather than sharing the stall phase's cap. And the exact-scheme solver still hard-wires the magnet-down field and had to be copied to measure the reversed-polarity ceiling; it should take the field as an argument, the way the derivative function and the reference integrator already do.
4. **Then, and only then, the continuous-time column on LHCb.** The decision to open it was parked until the residual redesign had been measured, because the redesign addressed the specific failure that would otherwise have been the reason to switch. The next slide records what the theory predicts there, before rather than after.

What the stability theory predicts for the continuous-time column on LHCb

Recorded before rather than after

- **There is no isolated fixed point to park on.** The right-hand side vanishes only where the slopes vanish and the transverse field does too, which for a track moving forward through the spectrometer is not a state it can sit at. The failure Rohrhofer et al. document, and that the regulariser was built to price, has no obvious analogue here.
- **Flow-following splices should be expected instead.** The splice construction needs only that the spliced-in piece solves the equation near the collocation points, and the system has a five-parameter family of solutions through every plane. A candidate that follows the true track for part of the interval and a neighbouring track thereafter has the same near-zero empirical residual. That is the family my regulariser arms converted parking *into*.
- **So the intervention to reach for is pseudo-time stepping with resampling, not the regulariser.** That ordering is a measured result over 240 converged runs, not a guess.
- **The difficulty axes are l_B and the chain length,** in place of the number of periods. A continuous-time study here should sweep those two and nothing else first.

How I would frame the whole thing in the thesis

The method is validated. The discrete-time construction transfers from a two-state autonomous oscillator with an analytic right-hand side to a five-component non-autonomous system with a measured, non-differentiable field. It reproduces the scheme it solves where the scheme limits; its shortfall where it does not is attributed rather than excused; and it works where no labels exist. Two systems with very little in common give the same loss-to-error rank correlation, +0.974 and +0.965, which I did not expect and cannot explain beyond the shared optimiser.

The product edge is unproven. No throughput measurement exists, the tails are not budgetable, and the one step the extrapolator exists for sits a factor 22 above the scheme it is solving with its supervised twin a factor 11. After two or three chained legs that single leg dominates everything else. The thesis can carry the first claim honestly today. The second is a set of experiments, not a set of words.

What I would like feedback on

1. **Is the throughput test worth doing now, or after the magnet leg is fixed?** It is the result that decides whether the programme has a product, but measuring the throughput of a network that is a factor 22 from its own ceiling may answer a question nobody will care about.
2. **Is a smooth field surrogate acceptable in principle?** It lifts the $23\text{ }\mu\text{m}$ ceiling for everyone and not only for me, but it changes the thing being solved. I would like to know whether that reads as a legitimate move or as a change of problem.
3. **What accuracy and what tail would actually be usable?** Every number here is scored against the scheme and against a straight line. I do not have a requirement to score against, and a 95th percentile target would change which experiments are worth running.
4. **Is the reversed-polarity result the right way to argue portability?** Labels are cheap here, so the claim I am making is that a loss which never touches a label is immune to changes of field map, geometry and convention. I would like that argument attacked before it goes into a thesis.
5. **How much of the van der Pol failure study belongs in the LHCb story?** It is what tells me which continuous-time intervention to reach for, and it is the only place the splice mechanism is measured rather than cited. But it is a different system and it may read as a detour.
6. **Is the stratified mix the right first move on leg B, or is it a distraction from more stages?** Both are cheap. I have argued myself into running the cross-magnet-only test first because its two outcomes point at different work, and I would like that reasoning checked.

Reference: the protocol and the gates, in full

The training protocol, unchanged since the July baseline

1. Build the dataset; transport training states exactly onto the leg's start plane if within 60 mm of it; compute the label-free scales from the resulting population.
2. Apply the fiducial requirement: discard any state whose reference trajectory leaves the field map.
3. Initialise a fully connected network with hyperbolic-tangent activations, double precision throughout, five inputs (seven for the general-leg experiments, carrying the start plane and the step length) and $4(q + 1)$ outputs.
4. Train with full-batch L-BFGS, 200 iterations per restart, strong Wolfe line search, history 120, single-threaded on one core.
5. **Stall**: restart until two consecutive restarts each improve the training loss by less than one percent.
6. **Confirm**: fresh optimiser, curvature history discarded; converged only if it re-stalls within two restarts without moving the train, validation or test medians by more than one percent.
7. Score on the held-out test split, which contains no particle that appears in training.

Three gates on the shared package, all passing

- **Vendoring parity.** The copied field-map loader returns bit-identical values to the original on 200,000 random in-map points: measured difference exactly 0.0 T, and the map file's checksum matches.
- **Differentiable-field parity.** The PyTorch field must agree with the numerical original to 10^{-12} T. Measured: 1.3×10^{-15} T on both polarities.
- **Baseline reproduction.** The generalised code with its generalisations switched off reproduces the July baseline bit for bit: the same dataset of 1979/2062/2018 states with identical arrays, the same initial parameters, the same losses and gradients, and the first optimiser restart landing on the same number to the last bit.

All experiment folders import one shared package, so a difference between two experiments is a difference in the experiment and never a drifted copy of the physics or the optimiser: field-map loader, tableau construction, reference integrator, differentiable field, network, losses, dataset builders, scorers and training driver.

Reference: provenance

Repository and commits

- 4fe2491 (5 September 2026): the stage sweep, the architecture grid, the reversed-polarity test, the general-leg network and the shared package. **Pushed.**
- 333edf7 (6 September 2026): the residual arm, its results, chains and verdict. **Local, pending push.**
- 29e9928 (6 September 2026): the second wave at 4×200 . **Local, pending push.**
- Van der Pol companion: 072677a (the chapter with the failure study folded in) and e56b168 (the 240-run axes study).

Sample and field maps

Official LHCb simulation, 2024 minimum-bias, detector description and conditions of 17 October 2023, five of eight files mirrored locally (19 GB), 200 events. 323,533 rows split by particle 258,857/32,389/32,287 with seed 20260718. Canonical field maps at both polarities, grid $81 \times 81 \times 146$ at 100 mm spacing spanning $(-4000, -4000, -500)$ to $(4000, 4000, 14000)$ mm, trilinear interpolation, checksums recorded in the paper's provenance appendix.

Compute

Nikhef HTCondor, all jobs single-threaded and double-precision on one core.

- Stage sweep: cluster **5781154** (24 runs), **5781162** (8 resumed).
- Architecture grid: **5781153** (120), continuations **5781167**, **5781442**, **5781445**, plus **5781358** (10 twin runs).
- General-leg, first wave: **5781158** (26) with seven resubmission clusters.
- General-leg, second wave: **5781443** (13 runs, 12 converged) with **5781566**.
- Reversed polarity: **5781156** (13) with **5781163** and **5781164**.
- Residual redesign: **5781469** (26 runs, submitted 5 September 2026 at 21:14, all 26 confirmed) with three continuation rounds.

Every figure in this deck is copied unmodified from the corresponding mini-paper's figure directory, which in turn copies it from the experiment folder that generated it, from committed result tables only. The slide-by-slide figure and number map is in `README.md` next to this source.

Reference: where the full record lives

The two papers

- *A label-free surrogate for track extrapolation in LHCb: the discrete-time physics-informed construction measured against the scheme it solves.* 38 pages, 13 figures, 16 tables. Every number traced to a committed result file.
- *Physics-informed neural networks on the van der Pol oscillator.* 48 pages, 37 figures. The van der Pol thesis chapter.

The five LHCb write-ups

- Overview, July to September: <https://app.notion.com/p/3d25d544b9d981bc9b6befc96d06dc4e>
- Stage-count sweep: <https://app.notion.com/p/3d25d544b9d981b3922bfbbbea66889f4>
- Network size and seed: <https://app.notion.com/p/3d25d544b9d98151ba5de91f44d99447>
- One network for all leg types: <https://app.notion.com/p/3d25d544b9d981759a84cf138afaa0f7>
- Reversed magnet polarity: <https://app.notion.com/p/3d25d544b9d981888b46e46af43b2928>

Code and committed result tables: <https://github.com/GeorgeWilliam1999/LHCb-Extrapolation>. Datasets and model weights are not committed; they regenerate from the scripts, the recorded seeds and the committed metadata. Every figure is regenerated by its experiment folder's plotting script from committed CSV tables only.